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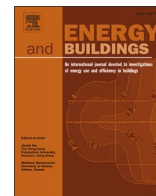
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Transfer learning in building dynamics prediction[☆]

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ABSTRACT

Buildings account for approximately 40% of global energy use, largely due to heating, ventilation, and air conditioning (HVAC) systems. Advanced control strategies, such as model predictive control (MPC), are crucial for optimizing energy usage in smart buildings. For effective MPC, high-fidelity models are necessary to reliably predict thermal responses under varying conditions. Low development costs, scalability and ability to process high-dimensional data and capture non-linear relationships make deep neural networks (DNNs) apt for predicting complex building dynamics. However, developing DNNs for new buildings is challenging due to their extensive data requirements. This paper investigates the use of transfer learning to address this issue, based on studies of different building models in different Norwegian climates. By pre-training control-oriented deep neural network models on synthetic building operation datasets, developed using EnergyPlus, from a similar source building in a similar climate, these models can be fine-tuned for new target buildings with minimal data. The study makes three key contributions. First, it introduces and evaluates six fine-tuning strategies for pre-trained DNNs, offering empirical insights into optimal approaches for adapting complex encoder-decoder architectures. Second, it employs two custom key performance indicators to quantify the effectiveness of transfer learning strategies, providing a standardized framework for assessing transfer learning in building dynamics prediction. Third, the study demonstrates the importance of fine-tuning specific model components, such as decoder layers, and the benefits of adding dense layers or gated recurrent units, especially in cold climates, where introducing colder weather data significantly improves predictive accuracy. Key findings include better results from fine-tuning decoder layers, especially early ones, and the importance of adding new layers before fine-tuning specific ones. Target buildings in mild climates and with similar physical properties (U-values) showed more stable and faster improvement from transfer learning. For the cold-climate case studies, introducing colder weather in the fine-tuning datasets significantly improved prediction accuracy.

1. Introduction

1.1. Background

Buildings are responsible for approximately 40 % of global energy demand and 36 % of greenhouse gas emissions [1]. Notably, nearly half of this energy use is attributed to heating, ventilation, and air conditioning (HVAC) systems [1]. Similar trends are observed in European countries [2,3]. Consequently, reducing energy needs in the building sector has become an imperative, driven by escalating environmental concerns, energy supply instabilities and rising energy prices.

The ever-increasing complexity of modern buildings necessitates

advanced building automation systems (BAS) for the efficient management of their various subsystems, such as HVAC, lighting, power distribution, safety, and security. Prior research supports that more intelligent control of these subsystems can substantially reduce energy use and thereby mitigate greenhouse gas emissions while maintaining or improving indoor comfort [4]. Contemporary building control mechanisms predominantly employ rudimentary rules-based controls with constant setpoints or stiff setpoint schedules, which offer limited energy-saving capabilities, suboptimal indoor comfort and occupant satisfaction [5,6]. A growing body of research advocates the efficacy of predictive control strategies, such as model predictive control (MPC), in substantially reducing building-related energy costs while optimizing

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comfort [7,8]. This assertion is substantiated through both simulation [9–13] and experimental studies [14–17].

A pivotal aspect of implementing advanced predictive control involves the development of a control-oriented dynamic model. This model must accurately predict indoor environment changes and calculate the impact of control actions such as HVAC temperature setpoints, HVAC modes (cooling, heating, heating and cooling, etc.), and window-opening signals on building dynamics, including indoor temperature and energy use. Importantly, the dynamic model should operate promptly [15,16], i.e., it must generate predictions of building variables for specific input time periods and conditions in a brief timeframe. This speed is crucial because a predictive controller may need to call upon the model hundreds to thousands of times at each controller step [7], depending on the search space of the control variables.

Like building numerical models, control-oriented dynamic models can generally be classified into three categories: white-box, grey-box, and black-box models [7]. White-box models, often referred to as physics-based models, demand comprehensive system information and rely on first-principle equations to calculate heat and mass transfer within a building. Programs like EnergyPlus [18] and IDA ICE [19] are commonly used to generate white-box models. Notable examples of control-oriented dynamic models in this category include CoBo for building controls [20], WFSim for wind farm control and simulation [21] and control-oriented models for chemical batteries [22,23].

Grey-box models, or reduced-order models, adopt simplified versions of the dynamic equations used in white-box models but rely on datasets for parameter identification [24]. A prevalent method in grey-box modelling is the use of resistance–capacitance (RC) networks based on the first principle of thermodynamics [25]. Examples of that include the use of RC control-oriented models for building control [24,26] and battery control [27]. Black-box models, also termed data-driven models, are constructed using statistical or machine-learning techniques. These models employ methods such as autoregression with exogenous inputs (ARX), autoregressive integrated moving average (ARIMA), autoregressive moving average with exogenous inputs (ARMAX), regression trees, support vector machines (SVMs), and deep neural networks (DNNs) to predict energy needs and indoor conditions [28]. Examples include the application of ARIMA for managing energy in electric vehicles [29], SVMs for mitigating diesel engine emissions [30], and DNNs for the automation of building systems [31] and flow estimation [32].

Reinforced by the massive amount of data induced by the deployment of metering and sensing technologies in buildings, the data-driven black-box approach has increased in popularity for building dynamics prediction in recent years [33]. These models offer the advantages of low development costs and scalability. The effectiveness of DNNs lies in their ability to process high-dimensional data and capture non-linear multivariable relationships, making them apt for intricate building dynamics prediction [34–36]. Prior studies have demonstrated that a sufficiently complex DNN can approximate any continuous function within a bounded set [37,38].

However, the application of DNNs often comes with the requirement of a large dataset for training to achieve sufficient performance. This challenge can be mitigated by leveraging transfer learning techniques. Transfer learning enables a DNN trained in one domain, often referred to as the source domain, to adapt its pre-learned features to a different but related target domain [39]. Transfer learning is a promising concept in computer science, inspired by the human ability to transfer and apply past knowledge to new tasks [40]. In recent years, various studies have explored the use of transfer learning methods to improve building energy prediction performance. Fan et al. [41] and Tian et al. [42] examined transfer learning strategies using open-source real measurement building datasets, respectively. Mocanu et al. [43] and Li et al. [44] also employed transfer learning methods to enhance prediction performance, each focusing on different building types. Ribeiro et al. [45] and Fang et al. [46] introduced specialized transfer learning techniques, such as the Hephaestus method and Long Short-Term Memory – Domain

Adversarial Neural Networks, to improve energy prediction accuracy. Most of these studies focuses on transfer learning using simpler models. Limited research exists on fine-tuning large, complex models, with most studies focusing on simpler architectures. Fine-tuning simpler models is significantly different from fine-tuning larger models because, in complex models, different layers can have vastly different effects on performance. In contrast, simpler models offer less flexibility and fewer components to optimize.

While the majority of studies have focused on using real building datasets, a few researchers have begun to explore the use of simulation-based synthetic datasets as a source for model pre-training, which can then be fine-tuned using smaller real-world datasets [47,48]. This approach aims to both reduce data acquisition costs and improve model performance, particularly when transfer learning methods are applied. Although considerable advancements have been made in the application of transfer learning for building energy prediction, a notable gap in the existing literature is the limited investigation into fine-tuning methods specific to DNNs, particularly those employing encoder-decoder architectures. Most existing studies have focused on traditional machine learning algorithms or simpler DNN architectures for transfer learning, largely overlooking the potential of advanced encoder-decoder-based DNNs in this context.

The contributions of this study lie in the development and evaluation of innovative strategies for transfer learning in building dynamics prediction using control-oriented deep neural networks (CONNn). The CONN architecture is a multi-step, non-recursive, multi-zone, and multi-output model that uniquely integrates both known past and forecasted future inputs. To the authors' knowledge, no prior studies have systematically explored hyperparameter tuning or evaluated fine-tuning strategies for DNNs of such depth and complexity in this domain. This research provides empirical evidence on how best to fine-tune such models, including a detailed analysis of the differential impacts of fine-tuning specific parts of the architecture, such as encoder versus decoder layers, as well as the effectiveness of adding new layers like dense layers or gated recurrent units (GRUs). These findings offer a comprehensive framework for optimizing DNNs for control-oriented tasks in data-sparse domains like building dynamics prediction.

The study employs a robust methodology that leverages synthetic datasets to assess fine-tuning strategies for CONNn in the context of transfer learning. The workflow involves pre-training CONN models on datasets from source buildings and systematically fine-tuning them with incrementally larger datasets from target buildings situated in different climates. This approach is applied across diverse building models and climatic conditions in Norway. A comprehensive and systematic parametric analysis is conducted to evaluate the impact of various hyperparameters and fine-tuning strategies on the performance of CONNn.

To quantify the success of these strategies, two custom Key Performance Indicators (KPIs) are introduced. The first measures the impact of knowledge transfer by comparing fine-tuned models with those trained on limited target data, while the second evaluates the improvement achieved through fine-tuning relative to pre-trained models. These KPIs provide a clear, quantifiable framework for measuring transfer learning effectiveness, enabling the identification of optimal strategies for adapting pre-trained CONN models to diverse building features and boundary conditions. By integrating novel methodologies, architectural advancements, and custom metrics, this study makes significant contributions to the field of transfer learning in building energy management.

2. Methods

2.1. Transfer learning evaluation workflow

In the domain of building energy use prediction, the efficiency and accuracy of neural network models are dependent on the training data, mainly its volume, its diversity, and its range of coverage (i.e., the

minimum and maximum extents of the dataset). A sizable dataset mitigates overfitting risks, ensuring the model does not specialize on the training set to the detriment of its generalization abilities. Diversity in the data safeguards against intrinsic model biases, ensuring the trained model is representative of the different scenarios it might encounter. Given this, the robustness of neural networks through diverse datasets becomes crucial when employing transfer learning, as it allows for the seamless adaptation of models from one building to another, despite differences in location and design.

In the context of building energy use prediction, transfer learning refers to the application of knowledge from a source domain (e.g., a specific building at a given location) to a distinct target domain (another building located elsewhere). Transfer learning encapsulates a myriad of techniques. While some approaches might involve adjusting all layers of the model, others might only fine-tune specific layers, leaving the rest fixed to retain previously acquired knowledge which is generalizable to all target domains. [49]. Additionally, new layers can be introduced to the pre-trained model and specifically tailored for the target task to optimize performance. In essence, transfer learning seeks to harness the robustness and knowledge encapsulated within models trained on a source domain with plentiful data, and adapt these models to target domains where data is sparse, thus potentially leading to more accurate predictions for the latter (see Fig. 1).

However, it is pivotal to recognize that the effectiveness of transfer learning depends on several factors. Both the strategy employed for model fine-tuning and the source-building dataset can influence the success of transfer learning in a given case. Additionally, the inherent complexity of building physics in both source and target buildings plays a crucial role. Differences in architectural design, building standards, occupancy patterns, and operational systems can introduce complexities that may necessitate tailored adjustments in the transfer learning process.

To effectively understand the potential of transfer learning within the domain of building energy use prediction, a recurrent evaluation process, wherein 29 CONN models with identical model architecture are trained, tuned and tested across diverse scenarios (see Fig. 2), has been employed throughout this research..

The framework is specifically designed to evaluate the impact of different aspects of transfer learning, including choice of training hyperparameters, choice of transfer learning strategy and combination of source and target building datasets. The 29 models comprise 3 base models and 26 fine-tuned models. The base models are as follows:

- **Pre-Trained Source Model (PT-SM):** Model trained using the complete dataset (full year) from the source building. It represents the accumulated knowledge derived solely from the source building and serves as a foundation for further fine-tuning with target data.

- **Baseline Target Model with Half Data (B-TM-half):** Model trained exclusively on the first half (weeks 1 to 26) of the dataset from the target building. It provides insight into how a CONN model performs when it is only exposed to a limited amount of target-building data.
- **Baseline Target Model with Full Data (B-TM-full):** Model trained on the complete dataset from the target building. This model serves as a performance benchmark.

In addition to the three base models, 26 fine-tuned models, each labelled as **Fine-Tuned Source Model $-i$ (FT-SM- i)**, are based on fine-tuning of PT-SM. Each of these models is fine-tuned with an incrementally increasing amount of data from the target building. The index i indicates the number of weeks of data used for the fine-tuning, spanning from 1 to 26:

- FT – SM – 1: Model fine-tuned using only data from week 26 from the target building.
- FT – SM – 2: Model fine-tuned using data from weeks 25 and 26 from the target building.
- FT – SM – 26: Model fine-tuned using data from weeks 1 through 26 from the target building.

All 29 models are tested using the second half of the target-building dataset (weeks 27 to 52), and their performance is assessed by computing the mean absolute error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|.$$

Here, y_i represents the actual value, $\hat{y}_i$ denotes the predicted value and n is the total number of data points. The MAE is a common model performance metric with straightforward interpretability and no over-penalization [28]. MAE determines the average error magnitude without magnifying individual errors. This stands in stark contrast to metrics like the coefficient of variation of the root mean square error (CVRMSE), the root mean square error (RMSE) and the mean square error (MSE), which all square the errors, thereby penalizing large deviations much more. Such an accentuation can give a skewed picture, especially when the data set contains outliers. Given the potential for undue penalization by other metrics and the clearer and more straightforward perspective offered by MAE, it was deemed the most appropriate metric for furthering this study.

2.2. Control-oriented deep neural network model

The control-oriented deep neural network (CONN) model used in this study is developed using the *pymodconn* package, a specialized package created for the construction of modular sequence-to-sequence CONN [50]. The CONN architecture is based on two data inputs:

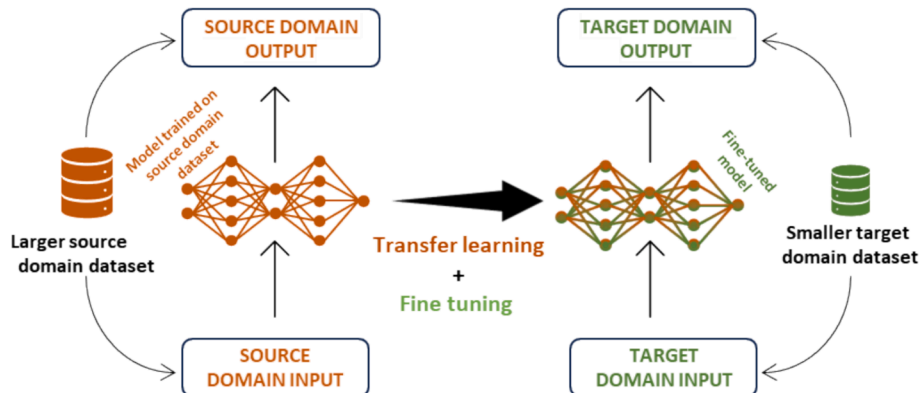


Fig. 1. Schematic of transfer learning principle.

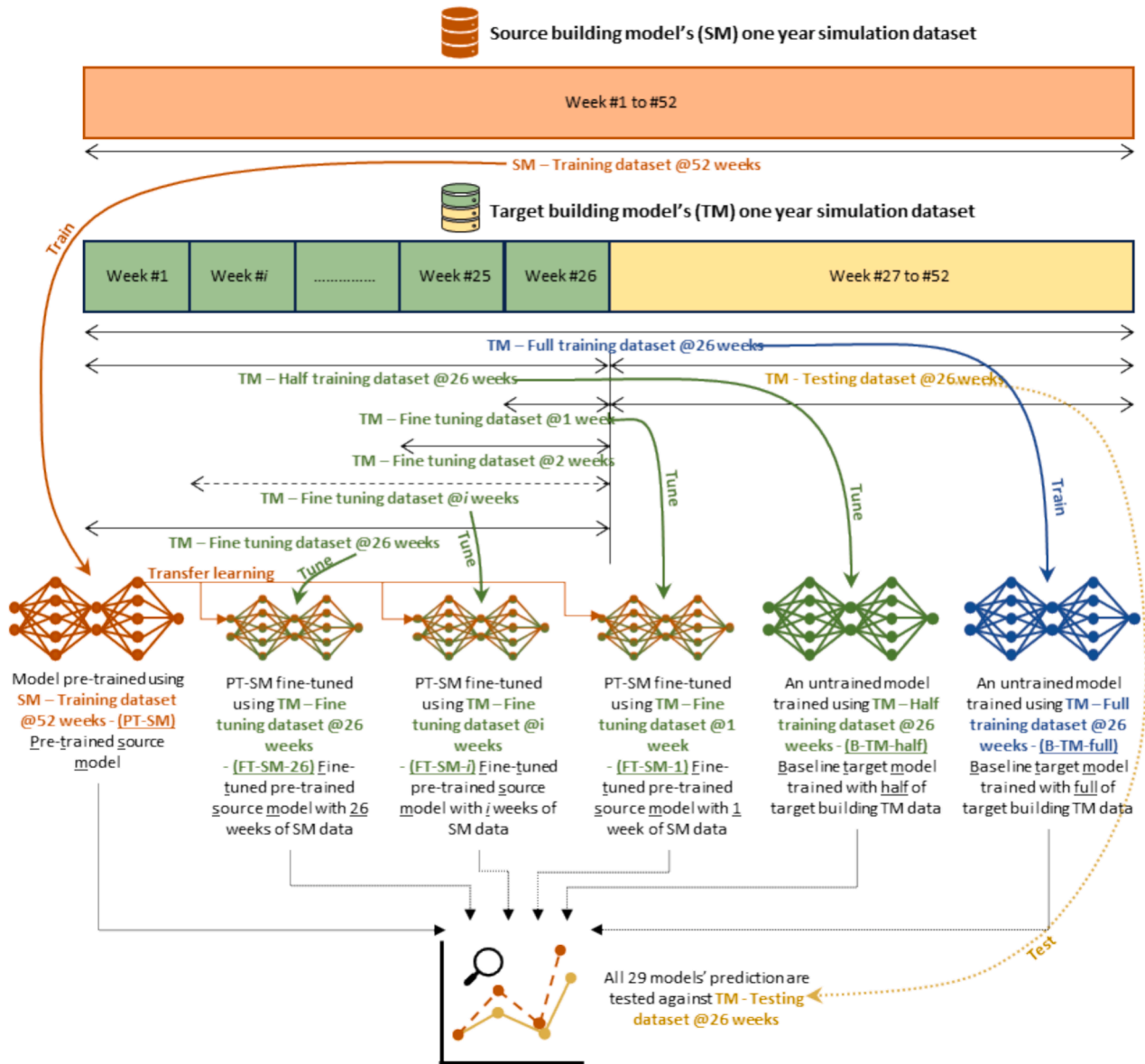


Fig. 2. Evaluation framework of this paper. Each iteration involves the fine-tuning and testing of a collection of 29 distinct models — Pre-Trained Source Model (PT-SM), Fine-Tuned Source Model – 1 (FT-SM-1) through FT-SM – 26, Baseline Target Model with Half Data (B-TM-half), and Baseline Target Model with Full Data (B-TM-full) — all against a common test dataset.

Known past data and *Known future data*. *Known past data* encapsulates the known historical information of a system, encompassing past system dynamics and previous control inputs. *Known future data* comprise target control inputs, time, known holidays or schedules, and forecasted weather data. The goal of a CONN is to synthesize these inputs and predict the *Unknown future data*, essentially projecting the future dynamics of the system.

The basic structure of a CONN developed using *pymodconn* consists of an encoder block and a decoder block, as illustrated for an example model in Fig. 3. The encoder block processes the *Known past data* and provides two outputs: a sequence of encoded states that carry the temporal intricacies of the past data and a final encoder state that offers a condensed representation of the entire past data. The decoder block is set up to ingest and process the outputs of the encoder block along with *Known future data* and predict the *Unknown future data*. The encoder block employs bi-directional gated recurrent units (GRUs) and self multi-head attention (MHA), whereas the decoder block has a

bidirectional GRU, self MHA followed by cross MHA, and another bidirectional GRU at the end. More details on CONNs and the model generation methodology can be found in the work by Chaudhary et al. [50].

2.3. Climate and building models

In order to ensure a representative evaluation covering different climates, five cities from different regions in Norway were selected: Tromsø in the north, Trondheim and Røros in the central regions, Bergen in the west, and Oslo in the south (see Fig. 4). All the chosen cities are characterized by climates dominated by a long heating period. A brief climatic overview of these cities, with average annual temperature (AAT) and heating degree days (HDD) for the year calculated using the weather files used in this paper, is given in Table 1.

Six distinct building models have been considered. The base model (M-0) is a small office building characterized by its compact structure

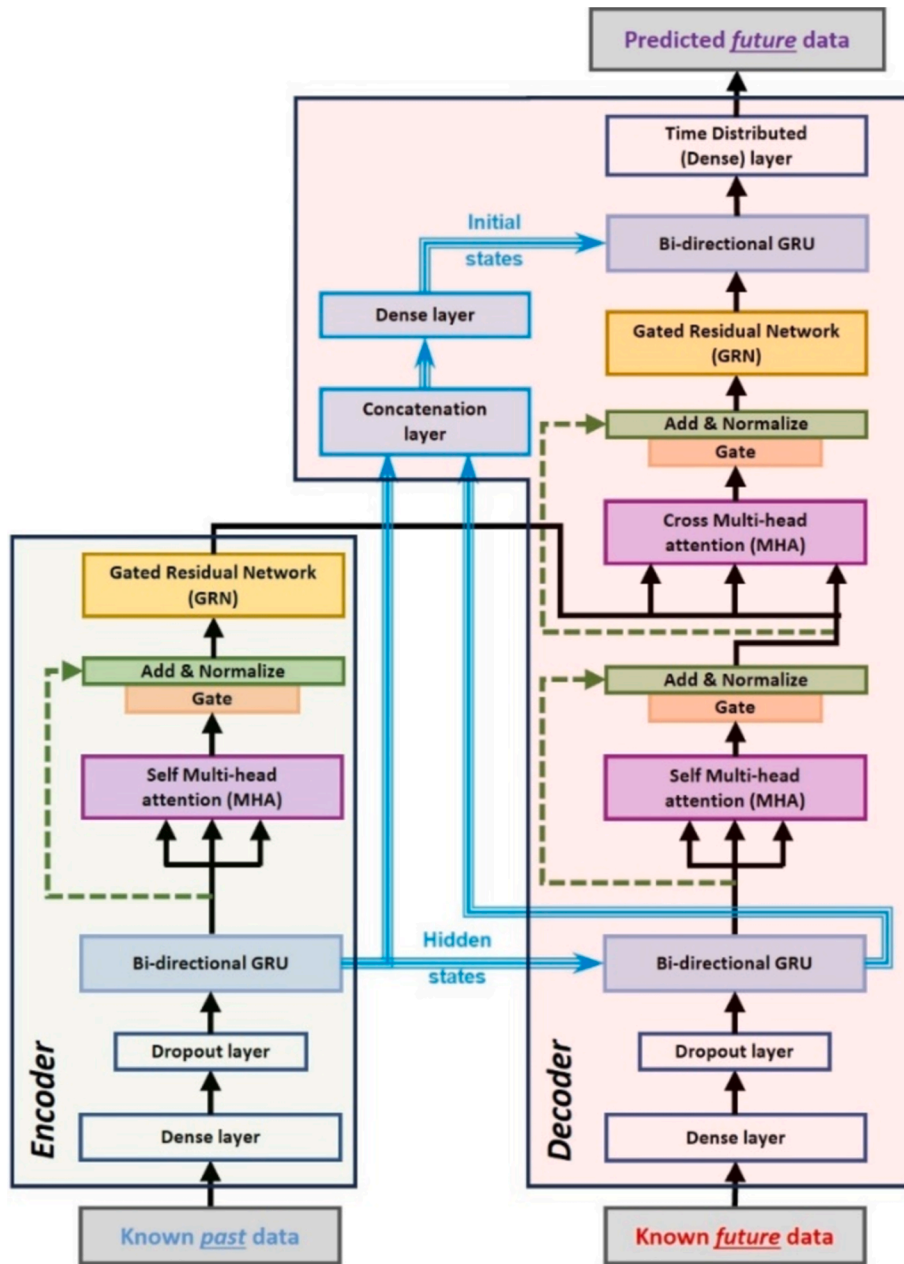


Fig. 3. CONN with bi-directional gated recurrent units (GRUs) and self multi-head attention (MHA) developed using the pymodconn package. This is the model architecture used in this study.

(see Fig. 5). With a single-floor design, the building spans 30 m x 15 m, with a ceiling at a height of 2.4 m. It is oriented 30 degrees East of North. The structure is partitioned into five distinct thermal zones, four exterior zones and one interior zone. Architectural highlights include windows across all four facades, glass doors situated on the southwest and northeast sides, and overhangs offering shade to the south-facing window and door. U-values for the different construction elements can be found in Table 2. Notably, there are no interconnections or openings between these zones. In this study, only the results for PERIMETER_ZN_1 will be presented.

Model M-1 is similar to model M-0 but with a different orientation. The building is aligned 90 degrees to the north, with PERIMETER_ZN_1 facing East. Model M-2 is also based on model M-0, but with a square layout (same overall floor surface area), zone height is increased to 3.2 m, no external shading, no doors, and window blind setpoint for incident solar radiation reduced to 35 W/m².

Model M-3 maintains the same architectural geometry as M-2, but has different envelope U-values. Model M-3 is further classified into three envelope efficiency categories: low, medium, and high. The low-efficiency variant aligns with buildings characterized by higher U-values, typical of less efficient structures, while the high-efficiency variant, with lower U-values, conforms to the TEK 17 [51] standard prevalent in Norway. The medium-efficiency variant has U-values that represent a balance between the low and high-efficiency standards.

2.4. Data description

The study employs *synconn_build*, an open-source Python package developed to produce synthetic data [52], which operates using EnergyPlus input data from the discussed building models, coupled with EnergyPlus weather files from the described cities. Random heating and cooling setpoint signals, randomized holiday lists, and stochastic

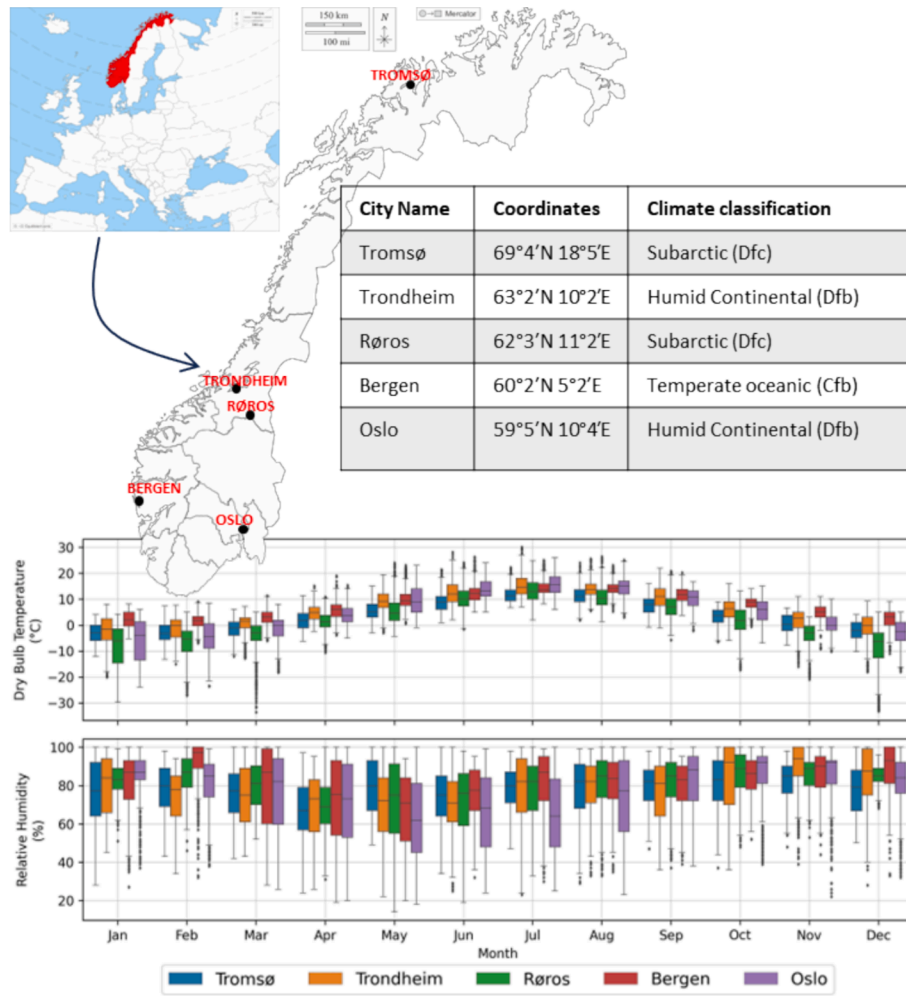


Fig. 4. Weather comparison for the five cities used for this study.

Table 1
Climate overview of all cities weather data used in this paper.

City	Description	AAT (°C)	HDD
Tromsø	North of the Arctic Circle, coastal location, high humidity	3.50	5314
Trondheim	Coastal location, high humidity	5.93	4441
Røros	Inland location, low humidity	1.44	6072
Bergen	Coastal location, high humidity	7.57	3837
Oslo	Coastal location, moderate humidity	4.85	4824

occupancy schedules for each building zone for each source or target model are generated by *syncomm.build*. More details about the data generation process can be found in the work by Chaudhary et al. [52]. A comprehensive list of the input and output variables can be found in Table 3. For this study, the CONN model has just one output, the total electric power demand in each zone for every timestep, encompassing lighting, equipment, and HVAC power demands, including heating and cooling.

The electric power demand in PERIMETER_ZN_1 from September to November for building M-0 in the Trondheim climate is presented in Fig. 6. This figure illustrates the daily and seasonal variations in electric power demand. The notable spikes in the graph are indicative of alterations in control inputs that impact power usage. Additionally, a weekly pattern emerges upon closer inspection. A span of three months is

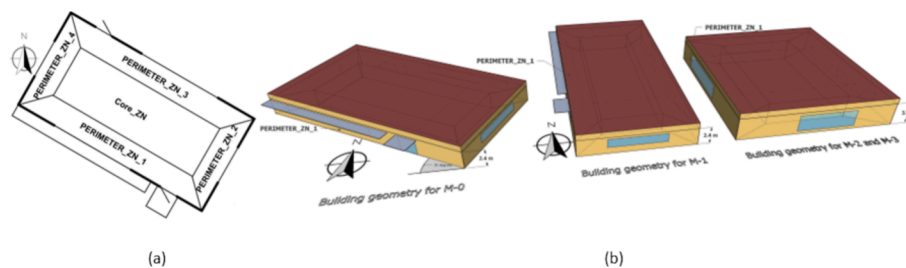


Fig. 5. (a) 2D plot showing zone structure for all building models, (b) 3D perspective of the three building geometries used, with PERIMETER_ZN_1 marked in each of them.

Table 2

Comparison of various building energy models developed for this study.

	M-0	M-1	M-2	M-3 (Low efficiency)	M-3 (Medium efficiency)	M-3 (High efficiency)
Orientation	30° to North	90° to North	90° to North	90° to North	90° to North	90° to North
Building shape	Rectangle	Rectangle	Square	Square	Square	Square
Zone height	2.4 m	2.4 m	3.2 m	3.2 m	3.2 m	3.2 m
Conditioned area	463.60 m ²	463.60 m ²	461.38 m ²	461.38 m ²	461.38 m ²	461.38 m ²
Conditioned volume	1132.80 m ³	1132.80 m ³	1476.43 m ³	1476.43 m ³	1476.43 m ³	1476.43 m ³
Window Shading	Shaded with louveres	Shaded with louveres	Louvres removed	Louvres removed	Louvres removed	Louvres removed
Window blinds setpoint for incident solar radiation	50 W/m ²	50 W/m ²	35 W/m ²	35 W/m ²	35 W/m ²	35 W/m ²
U-value – Exterior walls	1.48 W/m ² K	1.48 W/m ² K	1.48 W/m ² K	1.48 W/m ² K	0.60 W/m ² K	0.17 W/m ² K
U-value – Roof	0.27 W/m ² K	0.27 W/m ² K	0.27 W/m ² K	0.27 W/m ² K	0.27 W/m ² K	0.13 W/m ² K
U-value – Floor	2.64 W/m ² K	2.64 W/m ² K	2.64 W/m ² K	1.20 W/m ² K	0.63 W/m ² K	0.10 W/m ² K
U-value – Window	0.71 W/m ² K	0.71 W/m ² K	0.71 W/m ² K	2.56 W/m ² K	1.38 W/m ² K	0.71 W/m ² K

Table 3

Input and output variables used for the study.

Input variables (Known past data)		Units	Interval
T^{out}	Outside dry-bulb temperature	°C	[−35, 35]
H^{out}	Relative humidity of air	%	[0, 100]
W^{out}	Wind speed	m/s	[0, 30]
I_{Norm}^{out}	Direct normal radiation	W/m ²	[0, 1000]
I_{Hor}^{out}	Diffuse radiation on horizontal surface	W/m ²	[0, 1200]
h	Sine and cosine of the hour of the day	—	[−1, 1]
d	Sine and cosine of the day of the week	—	[−1, 1]
hol	Holiday	Boolean	[0, 1]
$Occu_i$	Occupancy in the zone	—	[0, 10]
E_i	Equipment load of the zone	W	[0, 1000]
L_i	Lighting load in the zone	W	[0, 1000]
HSP_i	Heating setpoint of zone	°C	[−35, 35]
CSP_i	Cooling setpoint of zone	°C	[−35, 35]
P_i	Total electric power demand in the zone	kW	[0, 7]
T^{out}	Forecasted outside dry-bulb temperature	°C	[−35, 35]
h	Sine and cosine of the hour of the day	—	[−1, 1]
d	Sine and cosine of the day of the week	—	[−1, 1]
hol	Holiday	Boolean	[0, 1]
HSP_i	Heating setpoint of zone	°C	[−35, 35]
CSP_i	Cooling setpoint of zone	°C	[−35, 35]
Output variables (Unknown future data)		Units	Interval
P_i	Total electric power demand in the zone	kW	[0, 7]

selected to effectively demonstrate the transitional trend in seasons, aligning with the expected changes in power demand.

A boxplot of the monthly power demand is shown in Fig. 7 for all building models specific to Trondheim. Each box represents the inter-quartile range of power demand for a particular month, with the horizontal line inside the box marking the median. Equivalent boxplots for the other cities can be found in the Appendix A as Figs. 21–24.

The heatmap shown in Fig. 8 illustrates the yearly energy usage for PERIMETER_ZN_1 across different building models and locations. The energy usage varies between the cities, likely due to differences in climate and weather patterns. Similarly, within the same city, the efficiency level of the building models affects the total energy use, with higher efficiency models generally using less energy. For instance,

model M-3 with low efficiency records the highest energy usage in Røros at 18.90 MWh, while the high efficiency M-3 model exhibits the lowest at 8.97 MWh in Bergen.

2.5. Transfer learning strategies

In this work, six different CONN transfer learning (TL) strategies have been investigated in order to identify what layers to adjust, re-train and add:

- TL-1:** The entire pre-trained model is fine-tuned. By adjusting all layers, the model aims to adapt its previously learned knowledge to the new target data.
- TL-2_x:** Only specific layers (denoted by x) are adjusted, while the rest of the model remains frozen. This provides targeted refinement, allowing the preservation of foundational knowledge while focusing on specific portions that might be more relevant to the new task.
- TL-3_x:** Specific layers (denoted by x) are re-trained from scratch (not just adjusted), while the rest of the model remains frozen. This enables the model to completely relearn aspects of the new task that may differ significantly from the source task, ensuring a more specialized adaptation. The layer numbers and group of layers are shown in Fig. 9.
- TL-4_y:** An additional layer (denoted by y) is introduced in the model and trained from scratch, while the rest of the model remains frozen. The introduction of fresh capacity helps capture unique features of the target data without disturbing the prior knowledge. The following layers were selected for testing:
 - Dense layer:** Recognizing the capacity of Dense layers to provide non-linear transformations and increase model complexity, dense layers with 25, 50, 100, 200 or 400 units were tested.
 - GRU:** Given their recurrent nature and potential for capturing additional temporal dependencies in the data, GRU layers with 25, 50, 100, 200 or 400 units were tested.
 - MHA:** Playing into the potential of attention mechanisms to weigh different parts of the input differently, MHA with 4, 8 or 16 heads were tested. Due to observed instabilities during the training of

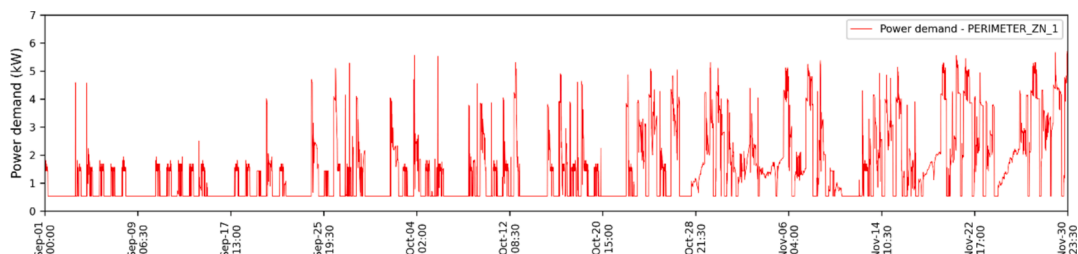


Fig. 6. Total electric power demand for PERIMETER_ZN_1 for time-period Sept–Nov for building model M-0 in Trondheim climate.

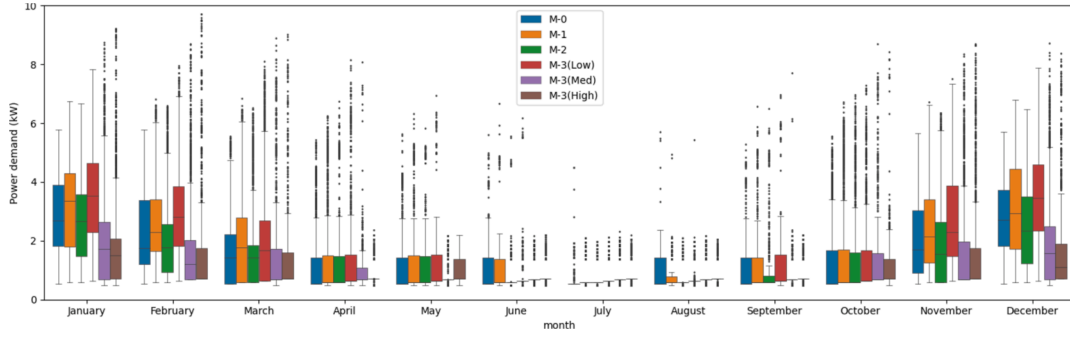


Fig. 7. Monthly distribution of power demand for all building models for Trondheim.

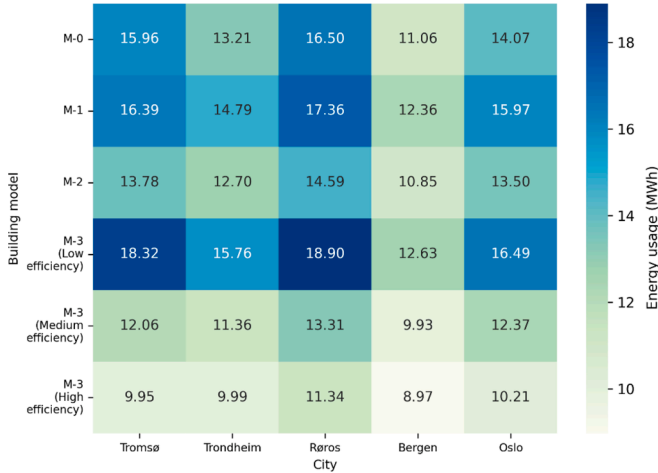


Fig. 8. Total yearly energy use in PERIMETER_ZN_1 for different building models in different cities.

the MHA layers, two separate runs were conducted for each configuration to reduce the influence of random variations.

5. **TL-5_x_y**: First, specific layer(s) x are fine-tuned, while the rest of the model remains untouched. Second, an additional layer 'y' is introduced and trained from scratch. This strategy aims to leverage both the refinement of existing layers and the addition of new capacities, thus increasing adaptability.
6. **TL-6_x_y**: First, an additional layer y is introduced and trained from scratch. Second, specific layer(s) x are fine-tuned, while the rest of the model remains untouched. By prioritizing the addition of new capacity before refining existing layers, the model can capture novel features early on and then refine its previously learned features for better synergy.

2.6. Transfer learning effectiveness

In the present study, two distinct key performance indicators (KPIs) based on MAE have been defined to gauge the transfer learning performance. The first KPI focuses on the effect of utilizing data from the source building, while the second KPI focuses on the effect of utilizing data from the target building for fine-tuning. In both cases, the MAE of the predictions from B-TM-full are used as the benchmark.

1. KPI_1^i – impact of knowledge transfer.

The performance of a model fine-tuned with i weeks of target data is compared with a model trained solely with half of the target building dataset:

$$KPI_1^i = 1 - \frac{(MAE_{FT-SM-i} - MAE_{B-TM-full})}{(MAE_{B-TM-half} - MAE_{B-TM-full})}$$

2. KPI_2^i – Impact of fine-tuning

The performance of a model fine-tuned with i weeks of target data is compared with the pre-trained model (with no fine-tuning):

$$KPI_2^i = 1 - \frac{(MAE_{FT-SM-i} - MAE_{B-TM-full})}{(MAE_{PT-SM} - MAE_{B-TM-full})}$$

The KPIs evaluate the merits of transfer learning from different perspectives. While KPI_1^i gauges the advantage of leveraging source data when target data is limited, KPI_2^i measures the incremental benefit of fine-tuning with target data. Ideally, using both KPIs in conjunction provides a better understanding of transfer learning's advantages under different scenarios.

KPI values below 0 indicate negative transfer learning, while values above 0 point to positive transfer learning [39,53]. In the context of KPI_1^i , positive transfer learning is observed when $MAE_{FT-SM-i}$ is closer to $MAE_{B-TM-half}$ than to $MAE_{B-TM-full}$. This means that the fine-tuned model, which leverages knowledge from the source building and is fine-tuned with 'i' weeks of target data, outperforms the baseline model trained only on half of the target building data. On the contrary, when $MAE_{FT-SM-i}$ aligns more closely with $MAE_{B-TM-half}$ than $MAE_{B-TM-full}$, the KPI value will be below 0. This indicates that the transferred knowledge may not have been beneficial and that a model trained merely on half of the target dataset might be more effective. In the context of KPI_2^i , positive transfer learning is observed when the $MAE_{FT-SM-i}$ is closer to $MAE_{B-TM-full}$ than to MAE_{PT-SM} . It suggests that fine-tuning a model initially trained on the source dataset with 'i' weeks of target data delivers better performance than the model trained only on the source building dataset. If $MAE_{FT-SM-i}$ is more aligned with MAE_{PT-SM} than with $MAE_{B-TM-full}$, the KPI will yield a value below 0. This underscores scenarios where the original knowledge from the source dataset, without any fine-tuning, might have been more beneficial for predictions on the target task.

2.7. Hyperparameter settings

This sub-section describes the training of the base models, namely PT-SM, B-TM-half, and B-TM-full for all source and target buildings. To enhance the performance of the CONN model, all model inputs and outputs are standardized within a uniform range $[-1.0, 1.0]$ using the *MinMaxScaler* Python function. Specifics concerning the minimum and maximum values of this scaling can be found in Table 3.

To set the study within the context of the MPC in buildings (typically requiring at least 24 h of ahead prediction), the model uses three days of past known data plus one day of future known data as input to predict 24 h (96 timesteps) ahead [7]. The time-series data is thus transformed into a supervised learning problem using the sliding window approach (see Fig. 10). The window length l represents the number of past time steps used to predict p timesteps into the future. Arrays $v_i(t)$ and $y_i(t)$ represent the input and output arrays at timestep t , respectively. For this study, the CONN model uses 'Known past data' from the time interval $[t$

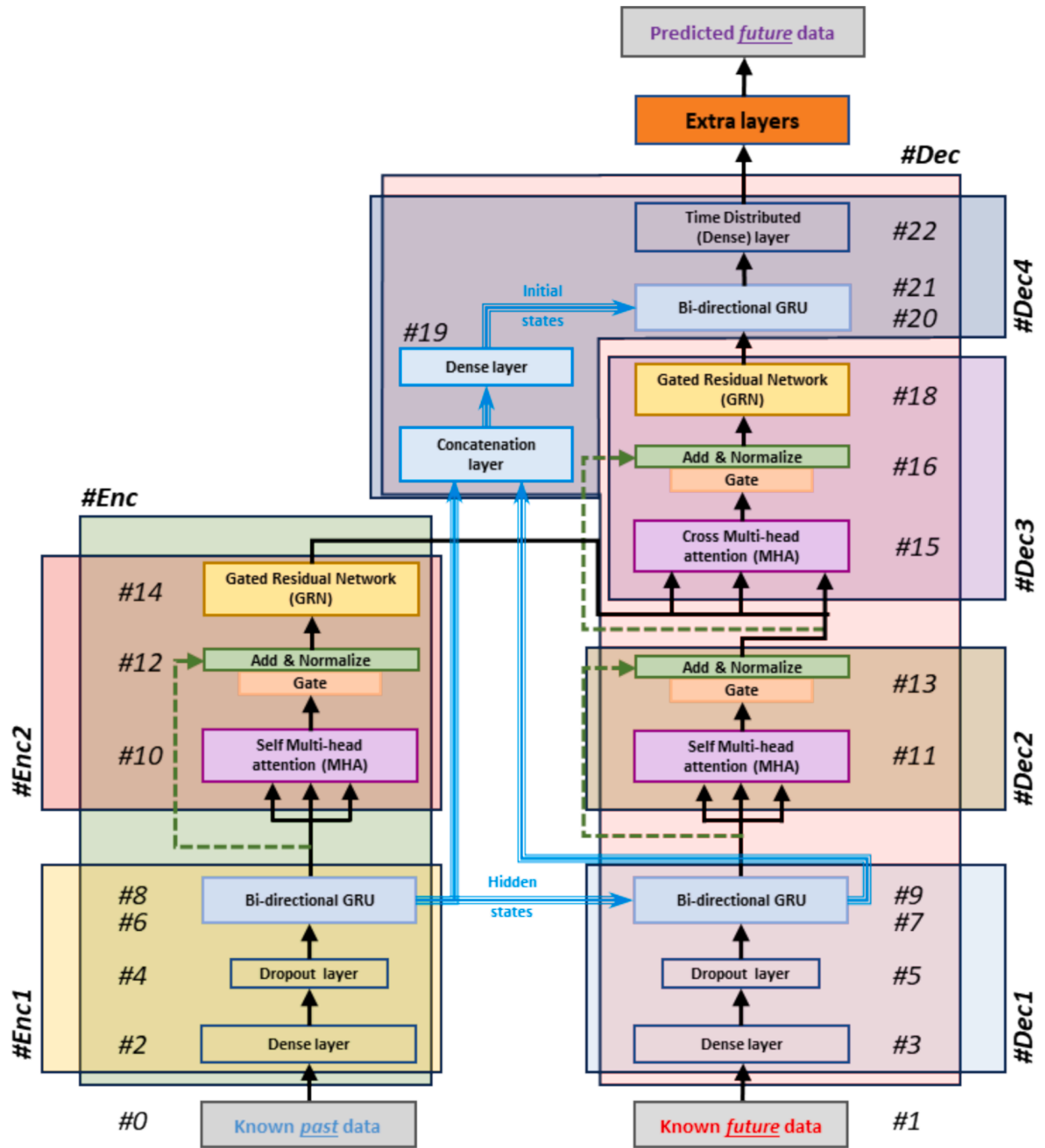


Fig. 9. Layer numbers and layer group names used for transfer learning.

– 287, t] and ‘Known future data’ from the time interval $[t + 1, t + 96]$ to predict ‘Unknown future data’ from the time interval $[t + 1, t + 96]$. Each timestep t is 15 min.

The CONN model is constructed using Tensorflow 2.11.0 [54] in tandem with the Keras library [55], all on the Python 3.10.0 platform. Hyperparameters utilized for the model’s configuration and its subsequent training are detailed in Table 4. The Adaptive Moment Estimation (Adam) optimizer [56] is used to minimize the neural network loss. This is paired with a cosine warmup restart (CWR) learning rate scheduler (see methodologies described in [57,58]). The CWR scheduler begins with a warmup phase to ensure convergence from a random initialization. The learning rate is then adjusted in a cosine curve, ensuring smoother convergence. In order to navigate away from local minima, intermittent warm restarts revert the learning rate to its initial value.

Values for the different hyperparameters were selected based on a case with M-0.Trondheim and M-2.Bergen as the source and target model, respectively, and TL-1 as the fine-tuning strategy. Batch size values of 128, 256 and 512 were tested. This decision was influenced by the technical advantages of NVIDIA GPU acceleration during model training, wherein batch sizes in multiples of 8 have been observed to be optimal. Regarding optimizers, Adam and stochastic gradient descent (SGD) were selected, given their representation of distinct optimizer families, ensuring a comprehensive spectrum for evaluation. Four different learning rates were tested: $1e - 3$, $1e - 4$, $1e - 5$ and $1e - 6$. A higher rate might lead to rapid convergence but risks overshooting the global minimum, whereas a smaller rate, although ensuring a more meticulous search, might become computationally expensive and get trapped in local minima.

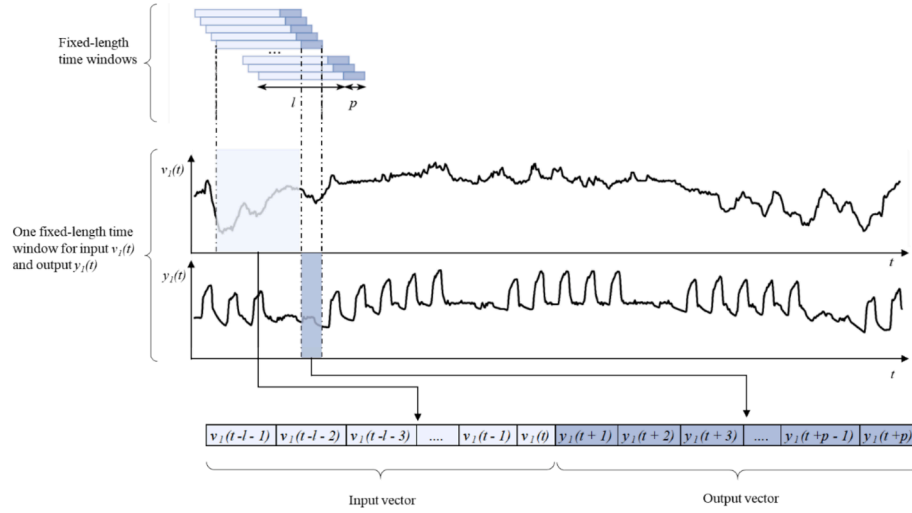


Fig. 10. Sliding window approach used to transform time-series data into a supervised learning problem, with example data not related to the study.

Table 4

Hyperparameters utilized for the configuration of the CONN models.

Hyperparameter name	Description	Value
n_{past}	Number of past data points for input data	288
n_{future}	Number of future input and output data points	96
ts	Number of data points in an hour (related to timestep)	4
CWR_{min_lr}	Minimum learning rate for cosine warmup restart (CWR) learning rate scheduler	0.0001
CWR_{max_lr}	Maximum learning rate for CWR learning rate scheduler	0.005
CWR_{cycle_length}	Cosine annealing length for CWR	6
CWR_{warmup_length}	Restart warmup length for CWR	3
$optimizer$	Optimizer used for training	Adam
$loss$	Loss function for the model	MSE

Beyond the MAE of the fine-tuned model, it is crucial to understand how much of the original model is retrained or modified in the fine-tuning. This is important to make sure the model stays generalized. If the performance degrades significantly on the source dataset but improves on the target dataset, the model might be overfitting to the target data. For a deeper understanding of this, the fine-tuned models are also evaluated against the source-building dataset, which is the original dataset on which the CONN model is pre-trained. The MAE for this test is here defined as the *self-test error*.

3. Results and discussion

3.1. Demonstration of prediction

Fig. 11 illustrates the prediction accuracy of a CONN model trained on the M-0 Trondheim model. The model was tested using simulation data from a city named Steinkjer, situated in close proximity to

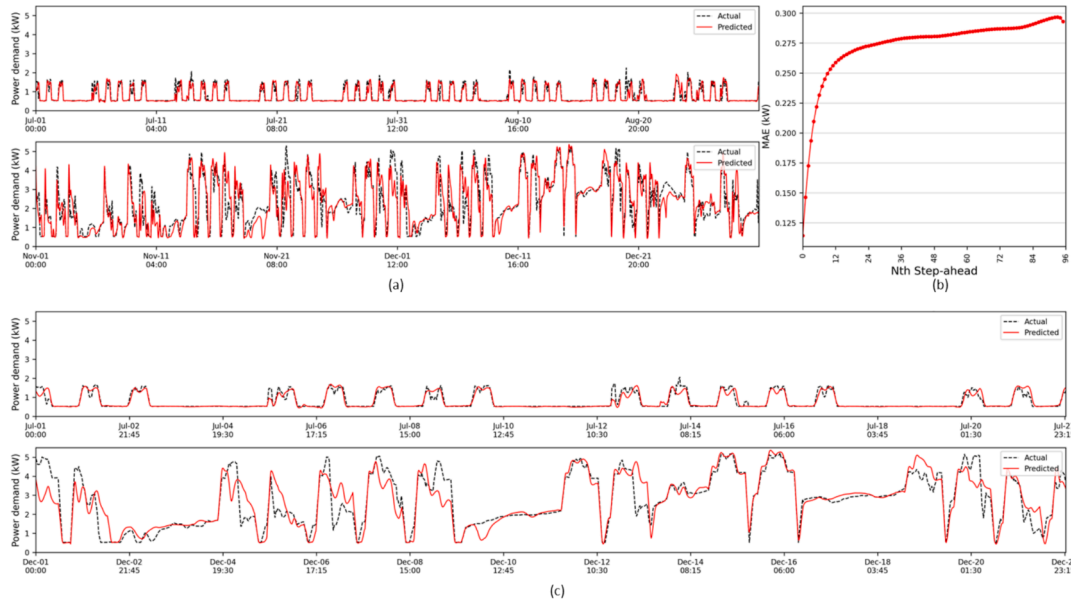


Fig. 11. Performance demonstration of a CONN with bi-directional GRUs and MHA, trained on full year dataset of M-0 Trondheim and tested on a six-month dataset of M-0 Steinkjer. (a) Actual vs. predicted total electric power demand for the 48th step-ahead prediction for July-Aug and Nov-Dec. (b) MAE (kW) for all 96 step-ahead predictions for the full testing period, (c) Actual vs. Predicted total electric power demand for the 48th step-ahead prediction for the first three weeks of July and December.

Trondheim, in order to have different weather but a similar climate. The selected testing duration spanned from week 27 to week 52, capturing both summer and winter conditions. While the model is projecting 96 steps into the future, the illustration predominantly focuses on the 48th-step prediction (i.e., 12 h into the future) juxtaposed with the actual data. Concurrently, the MAE of the prediction for each of the Nth step predictions is detailed. It is observed that the MAE for steps closer to the present is small, underscoring the relative ease of short-term forecasting. The MAE sharply rises until the 30th step, after which it exhibits a more gradual increase.

3.2. Hyperparameter settings

Results from the testing of hyperparameters, for the case with M-0_Trondheim as the source building, M-2_Bergen as the target building and TL-1 as the transfer learning strategy, are presented in Fig. 12. These plots shown from hereon show the average MAE for all prediction steps, i.e., 96, for 29 different models (as shown in section 2.1). The three subplots in Fig. 12 represent a specific batch size, and within each subplot, results are presented for the two optimizers (Adam and SGD) and the four learning rates. The continuous lines represent the target error, while the dotted lines denote the self-test error. The following observation can be made for the results:

- **Optimizer:** SGD demonstrates a slower rate of optimization, which can be attributed to its method of updating model parameters, using a fixed learning rate and the gradient of the loss function with respect to each parameter. This can lead to slow convergence, especially in complex models or data landscapes with many local minima, as SGD might get stuck or move very slowly around these areas. The Adam optimizer incorporates adaptive learning rates for each parameter by estimating the first and second moments of the gradients, which can lead to faster and more efficient convergence, especially in scenarios with large datasets and high-dimensional spaces. This adaptive nature is likely why, in this comparison, Adam outperforms SGD in terms of speed and efficiency, as observed in the smaller changes in

MAE for the models fine-tuned using SGD, both for target error and self-test error.

- **Learning rate:** A learning rate of $1e-6$ and $1e-5$ barely instigates any perceptible change in the model weights, as reflected in the negligible changes in MAE, suggesting an insufficient magnitude for the task at hand. Conversely, a learning rate of $1e-3$ seems to induce very large weight modifications, leading to large errors. Consequently, the $1e-4$ learning rate emerges as a balanced choice, facilitating necessary tuning without over-compensating.
- **Batch size:** The 128 batch size trails in terms of its tuning speed, being markedly more time-consuming than its counterparts. Batch sizes 256 and 512 are more comparable in their performance, with no substantial distinction in their efficacy. However, given the inherent benefits of larger batch sizes on the overall training duration, a batch size of 512 offers a strategic advantage in terms of computational efficiency.

To assess the repeatability and potential influence of randomness during training, the full evaluation was repeated five times for two distinct learning rates ($1e-3$ and $1e-4$) with the Adam optimizer and a batch size of 512. Detailed results of these evaluations are presented in Fig. 25 in the Appendix A. While the results associated with the learning rate of $1e-4$ showcased consistent and stable performance across all iterations, inconsistencies were observed for both target error and self-test error in the results associated with the learning rate of $1e-3$. This suggests that a learning rate of $1e-4$ offers a more controlled and predictable optimization trajectory, minimizing the undue variations that could be attributed to GPU-induced randomness. Based on these evaluations, the optimizer was set to Adam, the learning rate was chosen as $1e-4$ and the batch size was fixed at 512 for the rest of the work.

3.3. Transfer learning strategies

In this sub-section, the different transfer learning strategies defined in section 2.5 are compared, using M-0_Trondheim as the source and M-2_Bergen as the target. The results for TL-1, given by the blue line with circular markers in Fig. 12 (b), are used as the baseline for the

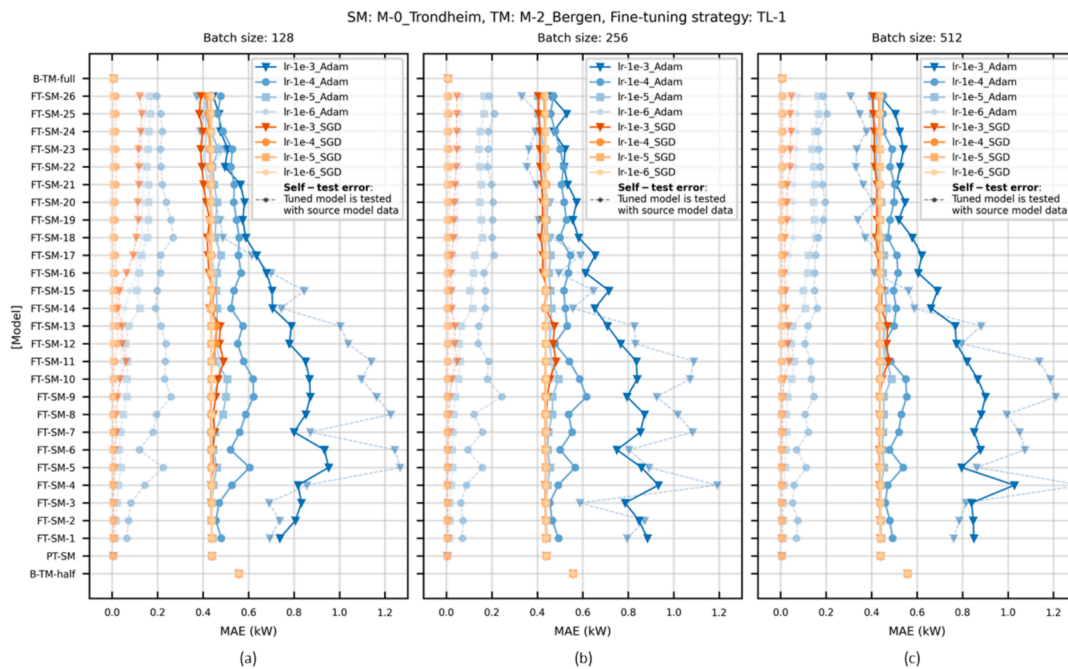


Fig. 12. Comparative evaluation of hyperparameter variations. This figure comprises three subplots, each representing a specific batch size: (a) 128, (b) 256, and (c) 512. Within each subplot, two different optimizers and four learning rates have been tested. The continuous lines represent the fine-tuning target error, while the dotted lines denote the self-test error.

comparison. In subsequent visual representations, the results for TL-1 are depicted as a black line complemented by circular markers.

Fig. 13 shows the results of both TL-2_x and TL-3_x for all trainable layers. Here, the individual layer is not important other than the type of layer, i.e., part of the encoder or decoder. Layers in the encoder are depicted in light blue, while layers in the decoder are coloured in light orange. For both strategies, fine-tuning of layers in the decoder consistently outperformed that of the encoder. This suggests that layers from the decoder might hold a more significant influence on the model's fine-tuning adaptability. In CONN models, including those with an encoder-decoder structure, it is widely acknowledged that the latter layers play a crucial role in fine-tuning [59–61]. The encoder part of the network generally captures the broad and fundamental patterns within the data, while the decoder is more adept at picking up on the finer, more detailed aspects specific to the task at hand.

Building upon the hypothesis that fine-tuning the later layers (or the decoder) has a more pronounced effect, a more expansive experiment was conducted. Instead of focusing on single layers, groups of layers were tested. This group could be the complete encoder, the complete decoder (or distinct subsections of the encoder and decoder, as defined in Fig. 9). As visualized in Fig. 14, the results confirm the hypothesis. Notably, fine-tuning the layer groups associated with the decoder demonstrated better performance than baseline TL-1, with the best performance observed when fine-tuning the layer group #Dec1.

In the context of encoder-decoder architectures, while the encoder is adept at compressing and representing information, the decoder is responsible for unravelling this condensed representation and generating a meaningful output. The first layers of the decoder are crucial, as this is where the transition from compressed representation to sequence generation (or prediction) takes place. Especially, considering the fact that the decoder is processing forecasted data in conjunction with control variables, these initial layers are likely central in determining the trajectory of system dynamics predictions, such as indoor air temperature or energy use.

In Fig. 15, the outcomes of the TL-4_x strategy are presented. A primary observation reveals that adding MHA layers, regardless of the

number of heads, leads to erratic results, especially for models fine-tuned with less than 15 weeks of data. Such behaviour is not surprising, as MHA structures are known for requiring extensive training datasets to stabilize and show their true potential.

In contrast, when Dense and GRU layers are added, better performance is observed. Dense layers can rapidly learn to identify and capture intricate patterns and relationships within the data, even with limited data. GRUs, with their ability to remember past data, can be especially proficient with time-series data. Their inherent design allows them to recognize patterns over shorter sequences, potentially making them more robust and adaptive even with limited data. The configurations Dense-200 and GRU-100 provided the best results and have been selected for deeper investigations in subsequent stages of the present study. Despite limited results, the MHA-8a configuration was retained for additional examination due to its distinctive structure and operational mechanics and called MHA-8 hereafter.

For the analysis of the transfer learning strategies TL-5_{x,y} and TL-6_{x,y}, all the decoder-related layer groups in Fig. 9 were tested for fine-tuning, while Dense-200, GRU-100 and MHA-8₁ were tested as additional layers, with the results of all combinations presented against the baseline TL-1 in Fig. 16. Analysing the results, it is evident that both strategies are most effective when #Dec1 is used for fine-tuning and Dense-200 is added as an additional layer. Subsequently, the pairing of #Dec1 and GRU-100 also exhibits commendable performance. TL-6 performed slightly better than TL-5, which can be attributed to the sequence of operations. Adding an extra layer and training it from scratch before fine-tuning existing layers provides the model with an initial adaptation to the new data. This preliminary adjustment potentially offers a more conducive environment for subsequent fine-tuning of specific layers, resulting in enhanced performance. Among the fine-tuning strategies tested, TL-6_{Dec1}Dense-200 performed best. This transfer learning strategy along with the best hyperparameters determined in the previous subsection (Adam optimizer, $1e-4$ learning rate and batch size at 512) will be used for all experiments in the next part of this paper.

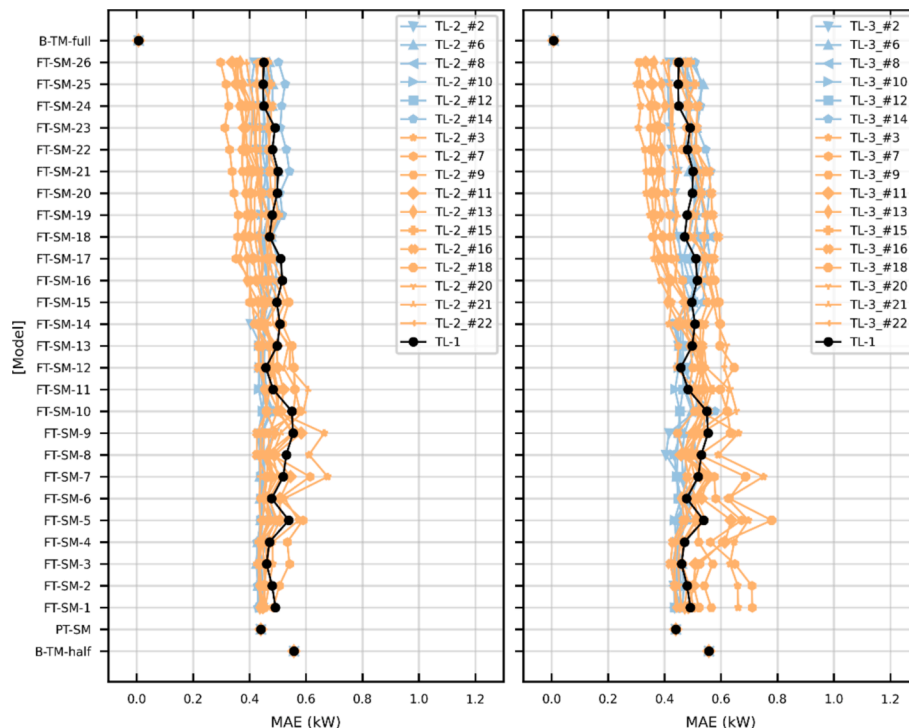


Fig. 13. Results of implementing transfer learning strategies TL-2_x and TL-3_x on specific layers. Layers in the encoder are depicted in light blue, while layers from the decoder are coloured in light orange. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

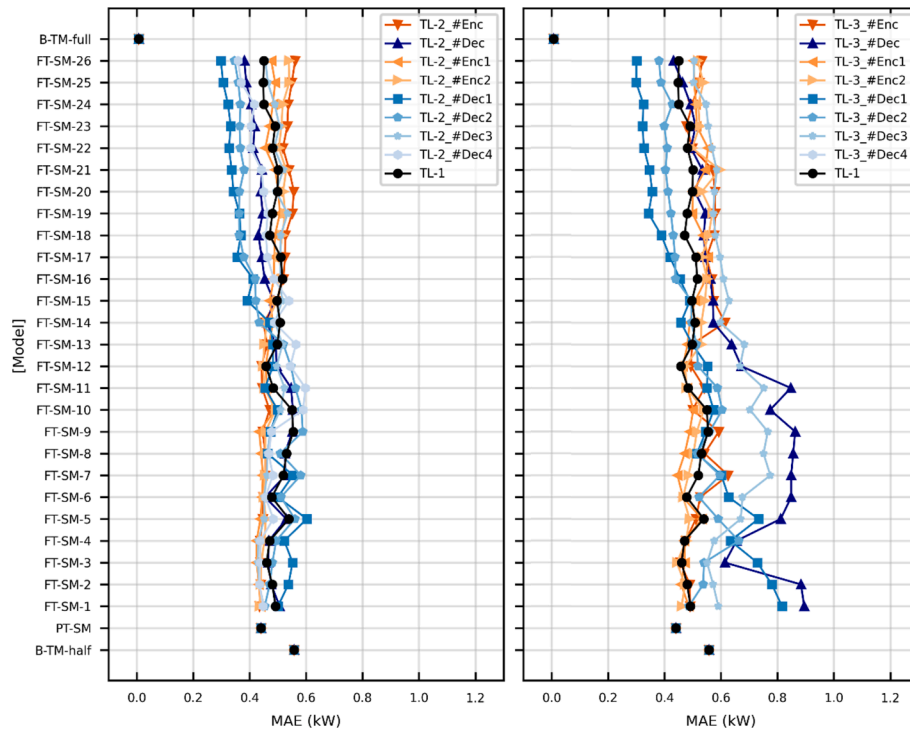


Fig. 14. Results of implementing transfer learning strategies TL-2_x and TL-3_x on a group of layers.

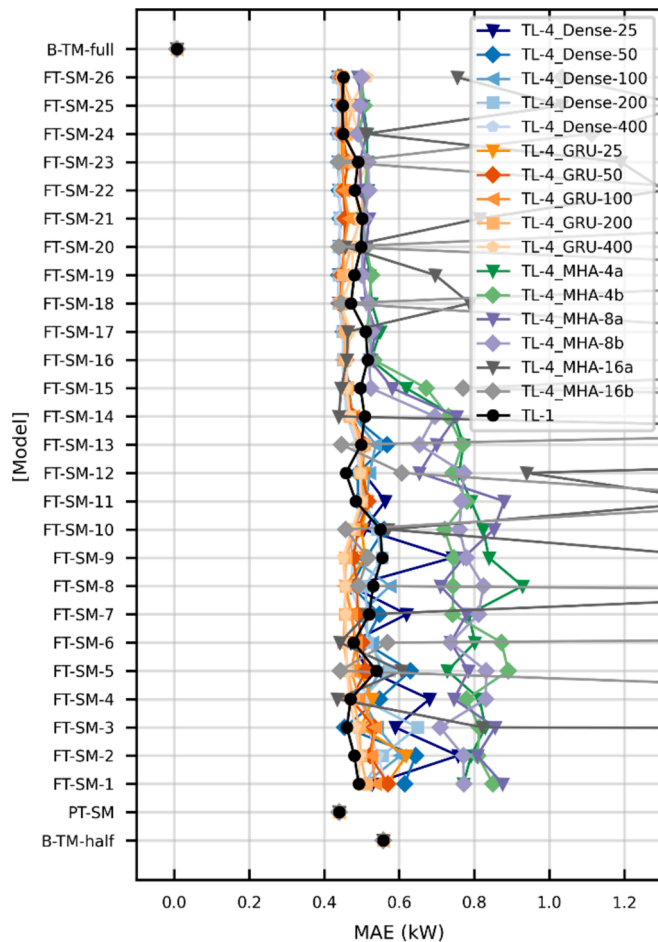


Fig. 15. Results of implementing transfer learning strategies TL-4_x.

3.4. Building model and climate

Building upon the hyperparameters and strategies ascertained from the previous sections, this subsection delves into a comprehensive transfer learning evaluation for various building models in different climates.

The voluminous results are synthesized and visualized using Level 5 plots, as outlined in [Appendix B](#). These plots, adept at presenting multi-dimensional data, encapsulate the performance of various combinations of source and target models on the two KPIs defined in [Section 2.6](#).

3.4.1. Power demand

In [Fig. 17](#), the mean absolute difference in total power demand for PERIMETER_ZN.1 is presented for different target buildings and target locations for the five different source model simulation data. Overall, the largest differences were observed with M-0_Tromsø and M-0_Røros as the source models, while the smallest differences were observed with M-0_Bergen as the source model, suggesting that fine-tuning would generally perform better with the latter as the source model. All building models vs city combination simulations (i.e., 30 cases) have different stochastic occupancy schedules, different sets of holiday dates and different profiles of heating/cooling setpoints.

3.4.2. Transfer learning analysis

Fine-tuning was performed for every pairing of source and target buildings. The results with M-0_Trondheim as the source model are presented in [Fig. 18](#), which includes six sub-plots for various building models, while equivalent figures for the four other source models can be found in the [Appendix A](#) as [Figs. 26-29](#).

The following observations can be made from the five figures:

1. For some combinations of target and source models, a smaller error is observed for the model trained with a full year of source model data than the model trained on six months of target model data, while for other combinations, the opposite is true. This is likely to depend on the relevance of the training data for the testing data, which varies with the building model and climate.

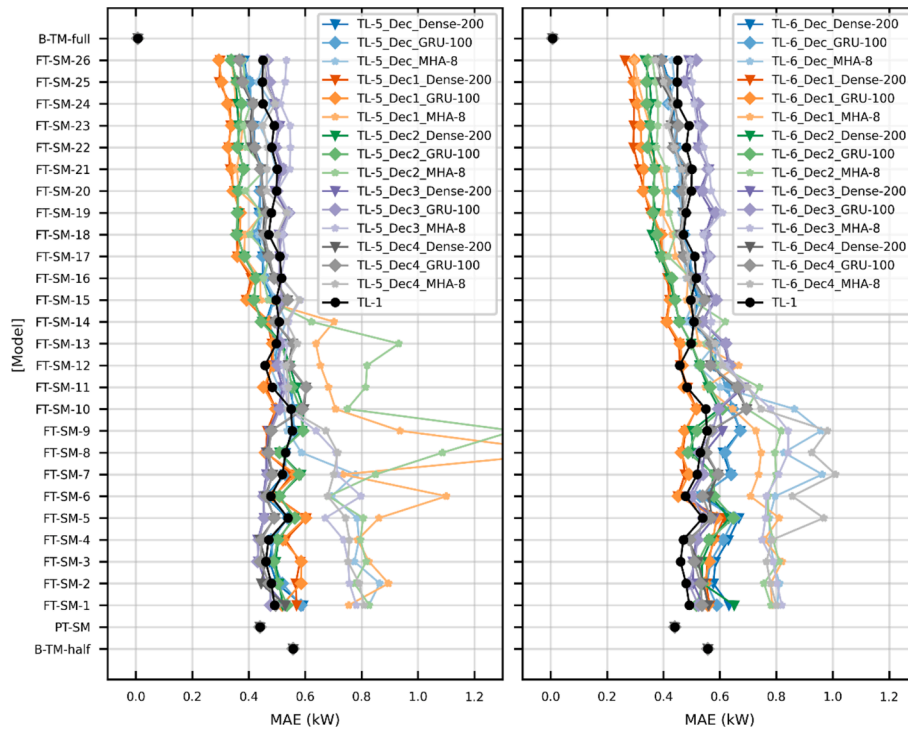


Fig. 16. Results of implementing transfer learning strategies TL-5_x_y and TL-6_x_y.

2. With limited fine-tuning data, the performance is worse than for the base model without fine-tuning, except for the cases where M-3 (High) is the target building and some cases where M-3 (Med) is the target building, for which improvement is observed already with one week of fine-tuning data. The performance of the fine-tuned model typically surpasses the performance of the base model without fine-tuning when more than 7–15 weeks (depending on the case) of fine-tuning data is used. With 26 weeks of fine-tuning data, the performance is better than both the base model without fine-tuning and the model trained solely with 26 weeks of target-building data.
3. Consistently, across all combinations of source and target models, the error reduces when colder weather data is introduced erroring the fine-tuning data, observed as a significant improvement in MAE from FT-SM-16. The explanation for this lies in the fact that colder weather introduces unique dynamics in building operations, such as higher heating demands, changes in energy usage patterns, and shifts in system behavior. If the model's pretraining does not include exposure to such conditions, it lacks the context to predict this behavior accurately. This highlights the importance of including representative data during fine-tuning.
4. Irrespective of climate, the largest errors are observed with M-3 (Low) as the target model, primarily because the building physics properties of this model differ the most from the base model.
5. Reflecting on the distinctions among different building models, M-1 deviates from M-0 solely in terms of orientation, causing PERIMETER_ZN_1 to shift from facing south-east to facing east. This change produces discernible results in transfer learning scenarios, especially when the target model is M-1_Tromsø, regardless of the source model chosen. The reason for this distinction becomes clearer when delving into the geographical and climatic nuances of Tromsø. Located at a higher latitude, Tromsø experiences longer durations of low-angle sunlight. This low-angle sunlight, especially during extended daylight periods, directly impacts east and west-facing zones of buildings. The direct eastward orientation of PERIMETER_ZN_1 leads to significant solar gain during sunset, altering the thermal dynamics and energy needs.

6. One fundamental takeaway from the evaluations is the profound influence of the climate. Røros, characterized by its colder and drier conditions, has a higher energy usage pattern than Bergen, which boasts milder and wetter conditions. The effect of this is seen in all combinations of source and target models, with the lowest MAEs observed for Bergen and the highest for Røros. However, it is imperative not to interpret lower MAEs as better transfer learning outcomes. A model with a broader range of power demand naturally exhibits higher MAEs and vice versa.

3.4.3. Transfer learning effectiveness

The results for the KPIs defined in section 2.6 are given in Figs. 19 and 20. Each figure is divided into five horizontal subplots, representing the five different source models. The KPI values are given on horizontal axes for the different target models on the vertical axes. For every source-target pairing, the KPI value is depicted by a circular marker. As the number of weeks of fine-tuning data increases (i.e., the value of i for model FT-SM- i), the size of the marker increases while the transparency decreases. Five distinct colours are employed, one for each of the five target model locations.

Within this section, various analyses derived from Figs. 19 and 20 are presented here, and Tables 5, 6 and 7 in the Appendix A are presented, structured as tables to capture the nuances of transfer learning dynamics across source-target model pairings. Table 5 showcases the range of KPI values, i.e., the highest and lowest KPI values for target cities grouped by source model city and target building model. The highest value is the right most whereas the smallest value is the left most. This table offers a comparative perspective on the efficacy of transfer learning by specific source models across various building dynamics and cities. Table 6 highlights the narrowest and broadest variability in KPI values for target cities grouped by source model city and target building model. The narrowest is the model which has the least highest KPI value minus the smallest KPI value and vice-versa. This analysis provides insights into the consistency or variability of transfer learning performance across different building models and climates when using specific source models. Table 7 showcases the range of KPI values variability, i.e., best 5 and worst 5, when categorized exclusively by the source model M-0 for

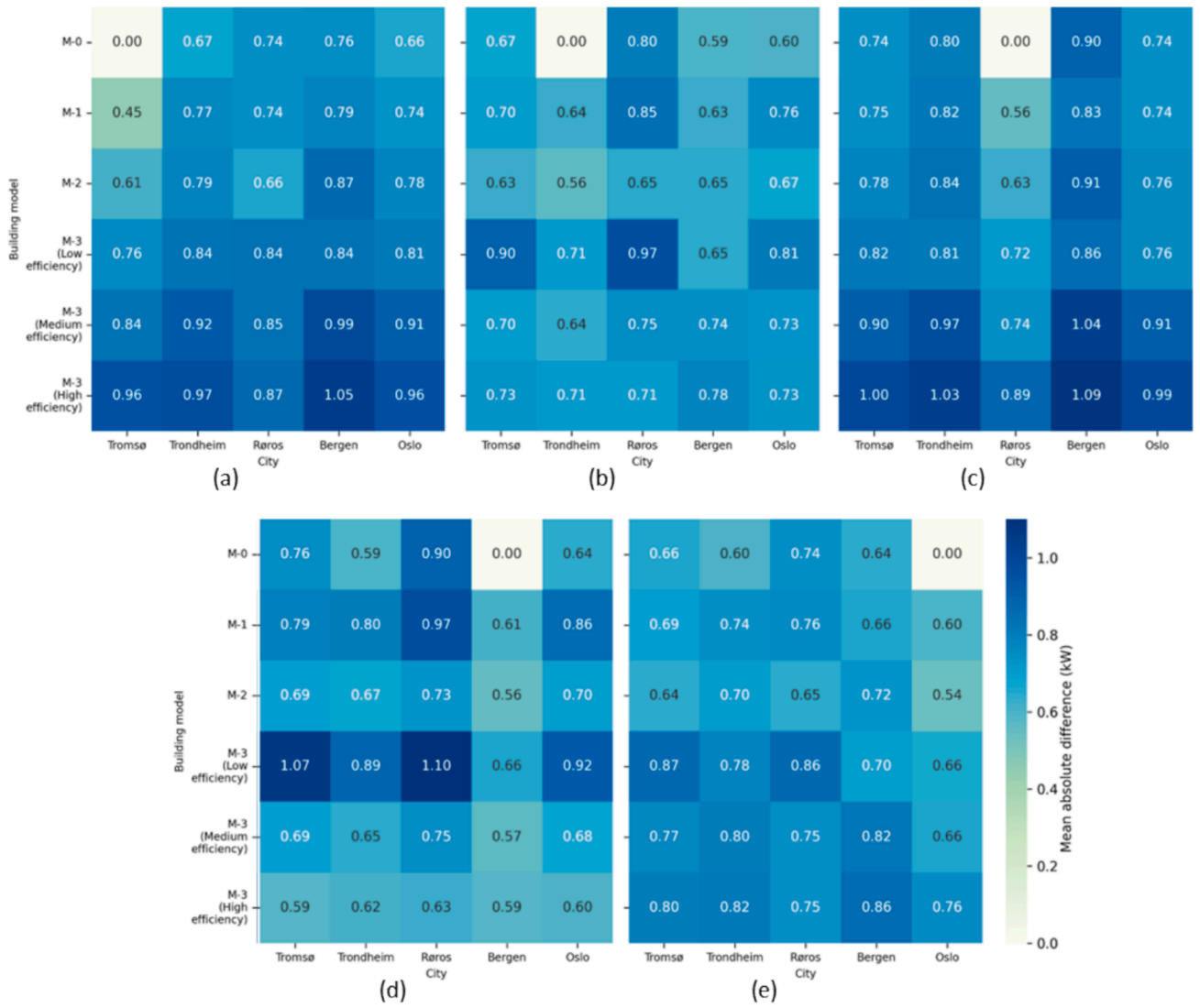


Fig. 17. Mean absolute difference in total power demand (kW) in PERIMETER_ZN_1 across multiple target models when compared to five different M-0 source models (a): M-0_Tromsø, (b): M-0_Trondheim, (c): M-0_Røros, (d): M-0_Bergen, (e): M-0_Oslo).

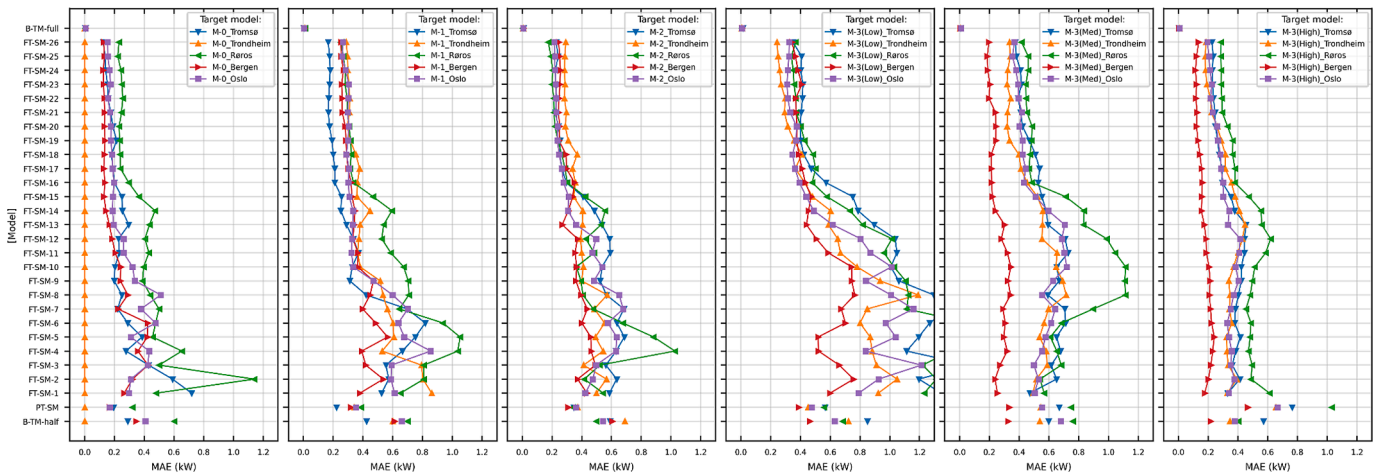


Fig. 18. Transfer learning results for all target-building models with M-0_Trondheim as the source building.

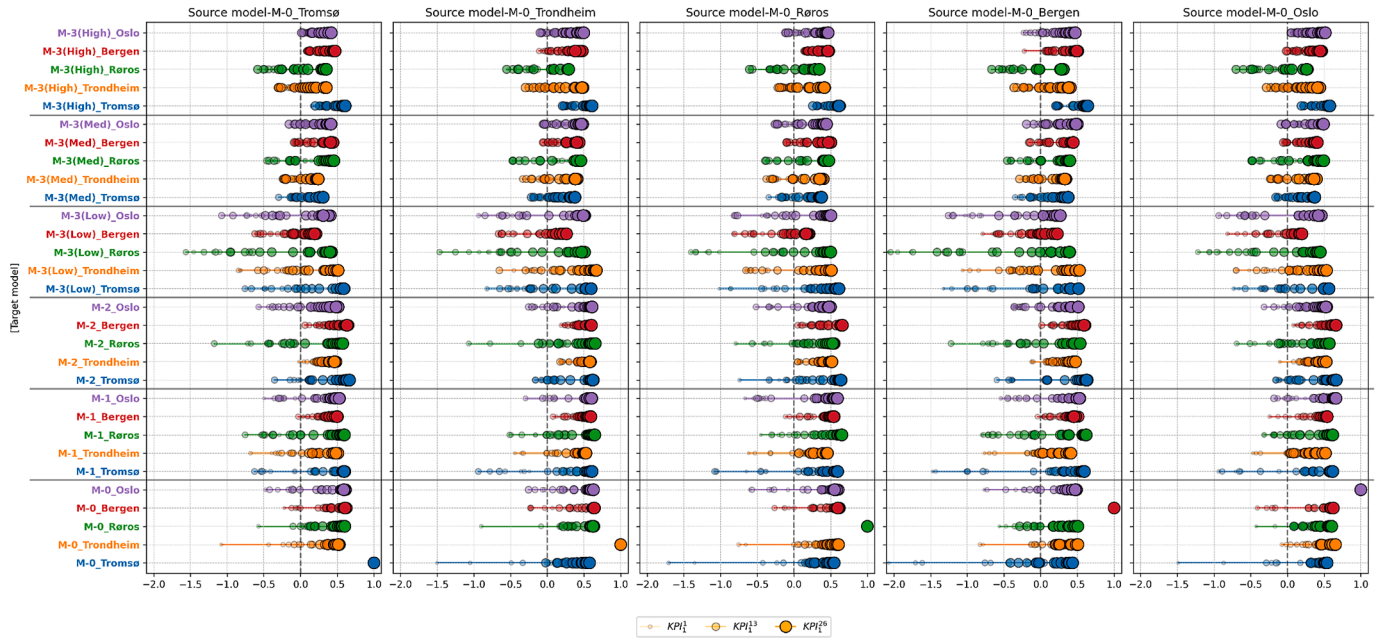


Fig. 19. KPI_1^i values for all combinations of source and target models, visualized across varying amounts of fine-tuning data. Larger, less transparent markers represent models fine-tuned with more weeks of target data.

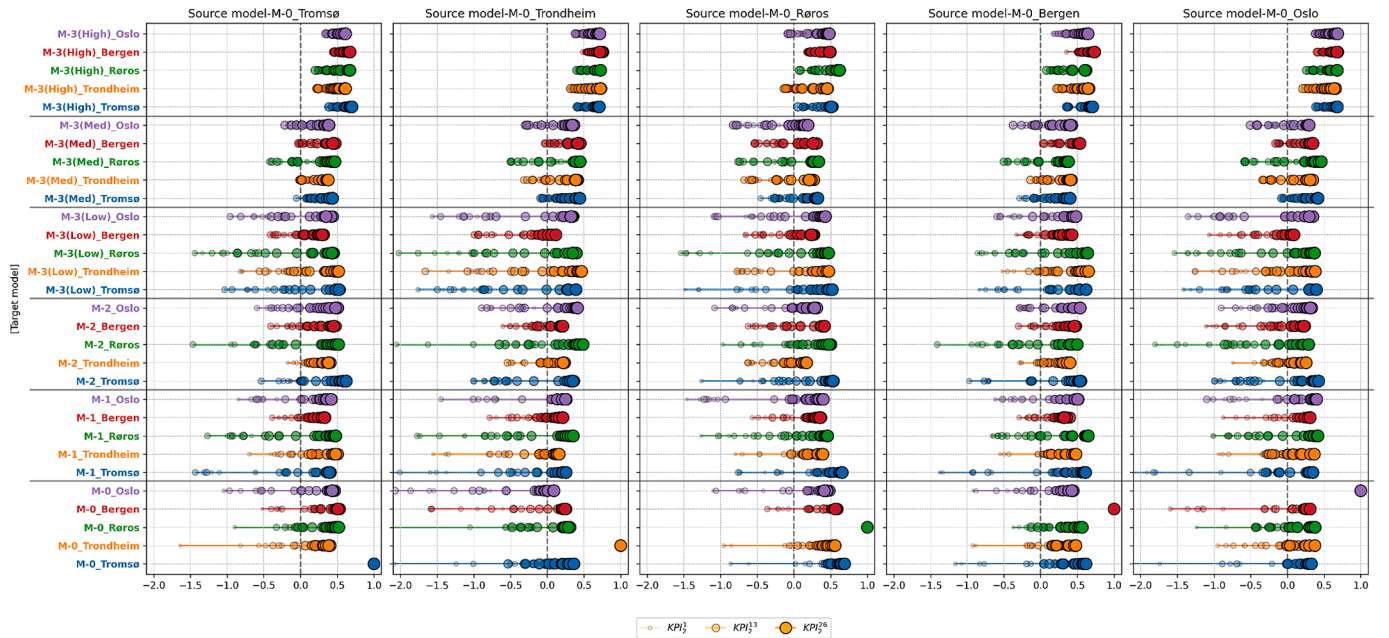


Fig. 20. KPI_2^i for all combinations of source and target models, illustrating the performance improvement due to fine-tuning. Larger, less transparent markers indicate models with more fine-tuning data.

each climate. This analysis provides insight into where each of the source models performs the best and worst among all target models.

The following observations can be made from both Figs. 19 and 20 and Tables 5, 6 and 7 from the Appendix A:

1. Overall, a KPI value range of 0.4–0.6 was seen for most of the cases, for all combinations of source and target models.
2. Across all source models, target models based on building model M–3 (High) showed the best values for KPI_2^i . This means M–3 (High) building dynamics was easier to learn using a model initially trained on a full year of data from model M–0. The range of KPI values was

also very narrow, meaning stable and consistent transfer learning effectiveness with incremental tuning data. A similar trend was not observed for KPI_1^i , meaning that the improvement seen with respect to models trained on half of the target model was not so significant. This could be due to the fact that the model trained on half of the target model data has already seen similar building dynamics, the model pre-trained on a full-year source model has not seen any of the target building dynamics before fine-tuning.

3. The smallest variability in both KPIs was consistently observed with Bergen as the target city. While the broadest variability was observed for Røros and Tromsø. A possible attribution for this distinction could

be climatic conditions. The harsher cold climates associated with Røros and Tromsø, as opposed to Bergen's mild and humid climate, might suggest that the former cities present more complex challenges for model tuning, while the latter offers a more predictable environment.

4. The smallest variability in KPI values was observed for M-3 (High), while the largest variability was observed for M-3 (Low) and M-1. This might be because there is less variation in the power demand for the more efficient building.

3.4.4. Selecting the optimal transfer learning model

The selection of the optimal transfer learning model is guided by the custom KPIs defined in Section 2.6:

- KPI_1^t evaluates the impact of knowledge transfer by comparing the performance of the fine-tuned model with a model trained on a smaller dataset from the target building. Positive KPI_1^t values indicate effective transfer of knowledge from the source model to the target model.
- KPI_2^t assesses the improvement due to fine-tuning by comparing the fine-tuned model's performance with the pre-trained model without any fine-tuning. Positive KPI_2^t values reflect significant enhancement from fine-tuning with target building data.

The optimal transfer learning model is identified as the one that demonstrates high values for both KPIs, indicating effective knowledge transfer and substantial performance improvement through fine-tuning. This balance ensures the model leverages the benefits of pre-training while adapting to the specific target domain.

In Figs. 19-20, the KPIs are visualized across different combinations of source-target models and varying amounts of fine-tuning data. Larger markers with higher transparency levels represent models fine-tuned with more weeks of data, which generally exhibit superior performance. Models that maximize KPI_1^t and KPI_2^t with minimal fine-tuning data are considered optimal, as they ensure computational efficiency without compromising predictive accuracy.

This selection strategy has practical implications for real-world scenarios, where obtaining large amounts of target building data may not always be feasible. By emphasizing both KPIs, we ensure the transfer learning approach remains scalable and effective under diverse conditions.

4. Conclusions

This study investigates the application of transfer learning to control-oriented deep neural network models for building dynamics prediction using synthetic datasets, with a specific focus on encoder-decoder architectures. Initially trained on a dataset from a particular building in a certain climate, the model is subsequently tested and fine-tuned on progressively larger datasets derived from a different building in a different location within Norway. The impact of different hyperparameters and fine-tuning strategies is examined. Two custom key performance indicators are proposed to gauge the success of transfer learning across diverse settings.

Key general takeaways from the study are listed below:

1. **Hyperparameters:** The Adam optimizer proved to be faster and more efficient than stochastic gradient descent for fine-tuning, with a learning rate of $1e-4$ providing a good balance between exploration and computational cost.
2. **Transfer learning strategy:** Fine-tuning the decoder layers is more effective than fine-tuning the encoder because the encoder primarily learns broad, generalized patterns from the input data, while the decoder handles task-specific outputs. The decoder layers are responsible for translating the encoded representations into

actionable predictions, making them critical for adapting the model to new target dynamics. Specifically, the early decoder layers play a pivotal role in processing forecasted data and control variables, which are central to predicting system behaviours like energy demand. By fine-tuning these layers, the model can better align its outputs with the nuanced requirements of the target domain, leading to improved accuracy and adaptability. Introducing an additional layer further improved the results. However, multi-head attention, requiring large amounts of data, was found to be less suitable than dense layers or gate recurrent units.

3. **Order of operations:** The best performance was observed when adding and training new layers before fine-tuning existing layers. The results suggest that the sequence of operations plays a critical role in effective model refinement.
4. **Climate:** The study focused on cold climates where building energy requirements are dominated by heating demands. When fine-tuning the models with incremental datasets, it was observed that the error decreased when colder weather data were introduced. Before being exposed to colder weather conditions, the models did not adequately understand the dynamics of building operations in such climates. Bergen, characterized by its mild and humid climate, consistently showed the smallest variability in fine-tuning performance. In contrast, Røros and Tromsø, known for their harsher, colder climates, exhibited the greatest variability. This suggests that the extreme conditions in these cities present more complex challenges for model tuning, requiring more data for fine-tuning to achieve comparable accuracy.
5. **Building dynamics:** Faster and more effective fine-tuning was observed for cases when the building source model had properties (U-values) close to those of the building target model, as the building dynamics in this case a more closely aligned.

The current study is based on extensive case studies requiring substantial computational effort. The key takeaways from the study, particularly the effects of climatic conditions on model performance and the impact of building dynamics on transfer learning, offer practical guidelines for similar research. Additionally, the insights into fine-tuning strategies, such as the importance of selecting appropriate hyperparameters and the order of operations in fine-tuning, provide valuable directions for transfer learning applications. These principles can be adapted and applied to different settings, allowing for the fine-tuning of pre-trained CONN models in new building and climate scenarios without necessitating the same level of resource intensity.

In real-time control applications, a pre-trained CONN model could significantly benefit from continuously receiving newly updated datasets from the target building, gradually improving the understanding of the building dynamics for the target building. Real-time application of these models would also facilitate the adoption of more advanced transfer learning approaches, such as the development of various ensemble models rather than relying on a single model. This approach would not only improve the robustness and reliability of the control system but would also allow for more dynamic adaptation to changing conditions, ensuring optimal performance in diverse operational environments.

Building on this, there is an opportunity to explore new paradigms for large-scale modeling that extend beyond individual tasks or specific building scenarios. Hybrid approaches that combine transfer learning with physics-informed principles could serve as a bridge to address data scarcity while enabling more universal applicability. Similar to reference building energy models, often referred to as stock models, a future direction could involve the development of very large CONNs with multiple decoders and encoders, trained en masse using highly refined synthetic datasets that capture a wide variety of building dynamics and climatic conditions. Once pretrained, these foundational models could be fine-tuned with real-world data from specific building operations, enabling them to adapt to the unique characteristics and operational

patterns of individual buildings. This approach would unlock the potential for deploying a single model architecture across multiple predictive tasks, such as energy forecasting, equipment performance monitoring, and predictive maintenance. By reducing the dependency on extensive data collection and model training for each specific task, this paradigm could significantly enhance scalability and efficiency, paving the way for more widespread adoption of intelligent building management systems. Such advancements would not only improve operational reliability and sustainability but also accelerate innovation in the domain of smart building technologies.

This study has several limitations and weaknesses which need to be discussed.

- First, the study only uses office buildings, which may limit the generalizability of the findings. Residential buildings, for instance, often have different conditions and more unseen factors such as varying occupancy patterns, diverse energy usage behaviours, and different HVAC requirements. These differences could significantly impact the effectiveness of transfer learning models when applied to residential settings. Long-term model drift is another critical issue. Over time, the dynamics of buildings can change due to factors such as renovations, changes in usage patterns, or the natural ageing of infrastructure. These changes can alter energy consumption patterns, necessitating continuous updates and retraining of the models to maintain accuracy. This ongoing maintenance is resource-intensive and can pose significant practical challenges.
- This study's pre-trained models could benefit from being trained on datasets from multiple climates to enhance their robustness and generalizability. However, this approach would require significantly more computational resources and data, making it more complex and costly. Such extensive training is necessary to ensure that the models can accurately predict energy dynamics across diverse climatic conditions.
- One significant limitation of this study is the absence of validation using real-world datasets. While the use of synthetic datasets provides a controlled and scalable means for pretraining and fine-tuning, it falls short in capturing the inherent complexities and variability of actual building operations. Incorporating real-world data from target buildings during the fine-tuning and validation phases would significantly enhance the model's applicability and accuracy. Validation with operational data would ensure that the model can effectively address real-world challenges and variability, bridging the gap between simulated conditions and practical deployment scenarios. This crucial step is necessary to establish the robustness and reliability of the proposed transfer learning approach in real-world applications.
- A notable limitation of this study is the absence of a direct performance comparison between the fine-tuned CONNs and traditional control-oriented models, such as Conventional State Space Models (CSSMs) and Resistance-Capacitance (RC) grey-box models. While such a comparison could provide valuable insights into the relative strengths and limitations of these approaches, conducting it within the scope of this work was not feasible. A fair and robust evaluation would require extensive multi-field expertise, encompassing control theory, system identification, and machine learning, to ensure a balanced and meaningful comparison. Future research could address this gap by replicating similar experiments with these traditional models to explore their performance under similar conditions. This would help determine the specific scenarios where CONNs demonstrate clear advantages or where traditional models remain more appropriate for building dynamics prediction.

- Finally, the finetuning process could be improved by incorporating techniques such as Reinforcement Learning from Human Feedback (RLHF) combined with layer-specific finetuning. This approach could help the models adapt more effectively to new data by leveraging human expertise and focusing on the most impactful layers, thereby enhancing the overall performance and adaptability of the transfer learning models. Integrating RLHF could address some of the limitations related to unseen factors and model drift, but it also introduces additional computational and methodological complexity.

This research offers significant potential for extended applications in the domain of smart building management and beyond. By demonstrating the efficacy of transfer learning for CONNs, the study provides a foundation for scalable, cost-effective approaches to predictive modeling for diverse building types and climatic conditions. These models can be employed to optimize energy management in real-time, enabling the integration of advanced control strategies, such as MPC, with reduced data requirements. The approach can also facilitate rapid deployment in new buildings, minimizing the need for extensive, site-specific data collection and training, which is often a barrier to adopting advanced control technologies in practice.

Beyond building energy management, the methodology and findings of this study can be extended to other fields requiring predictive control of dynamic systems, such as industrial process control, renewable energy systems, and smart grid optimization. For instance, the transfer learning framework could be adapted to predict and manage energy flows in microgrids or optimize operations in HVAC systems for non-building applications. Additionally, the ability to fine-tune pre-trained models with minimal additional data has implications for reducing computational costs and increasing the adaptability of AI-driven solutions in dynamic, resource-constrained environments. This could enable the development of intelligent systems capable of adapting to varying operational contexts, thereby promoting energy efficiency and sustainability across multiple sectors.

CRediT authorship contribution statement

Gaurav Chaudhary: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hicham Johra:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization. **Laurent Georges:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization. **Bjørn Austbø:** Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

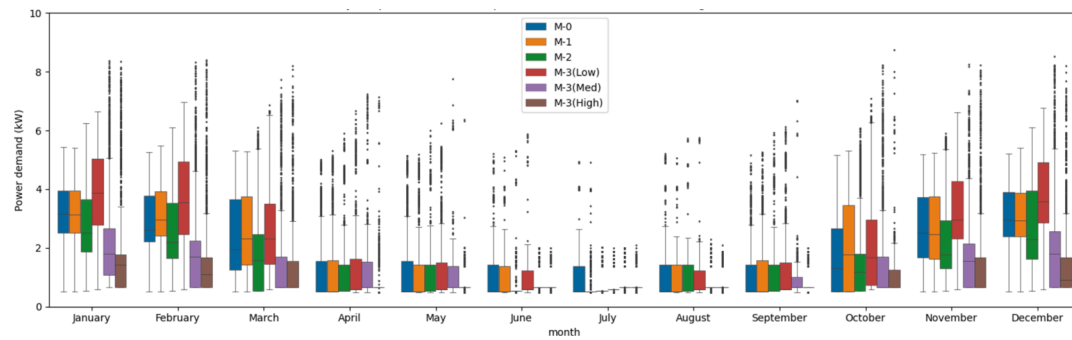


Fig. 21. Monthly boxplot distribution of power demand (kW) for all building models for Tromsø

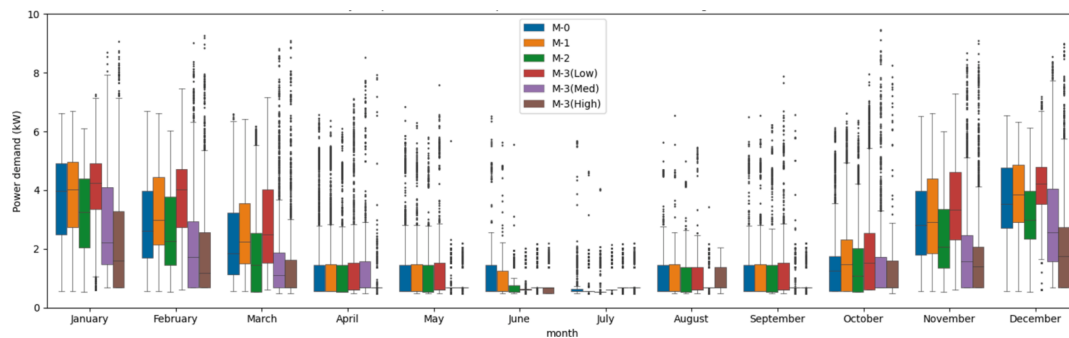


Fig. 22. Monthly boxplot distribution of power demand (kW) for all building models for Røros

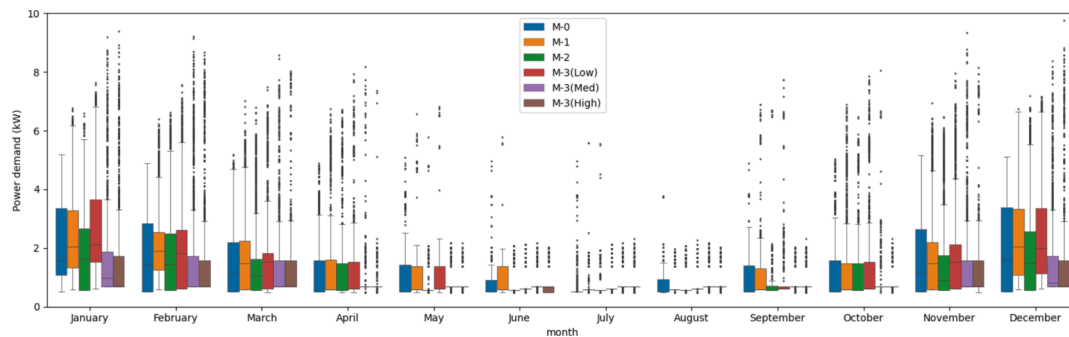


Fig. 23. Monthly boxplot distribution of power demand (kW) for all building models for Bergen

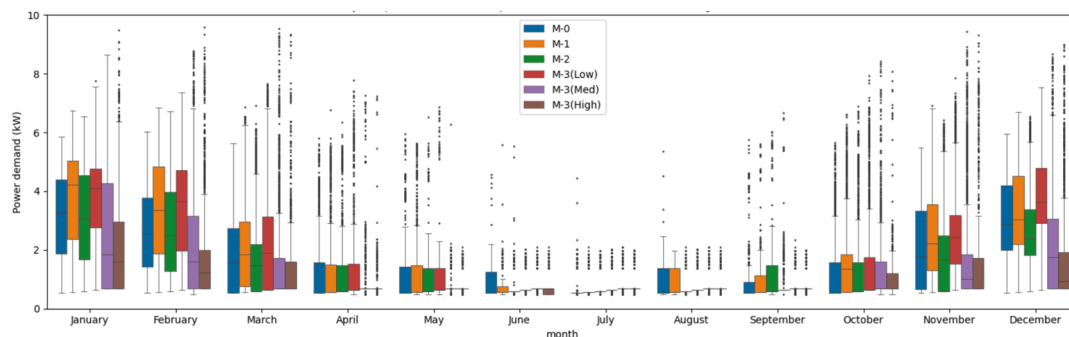


Fig. 24. Monthly boxplot distribution of power demand (kW) for all building models for Oslo

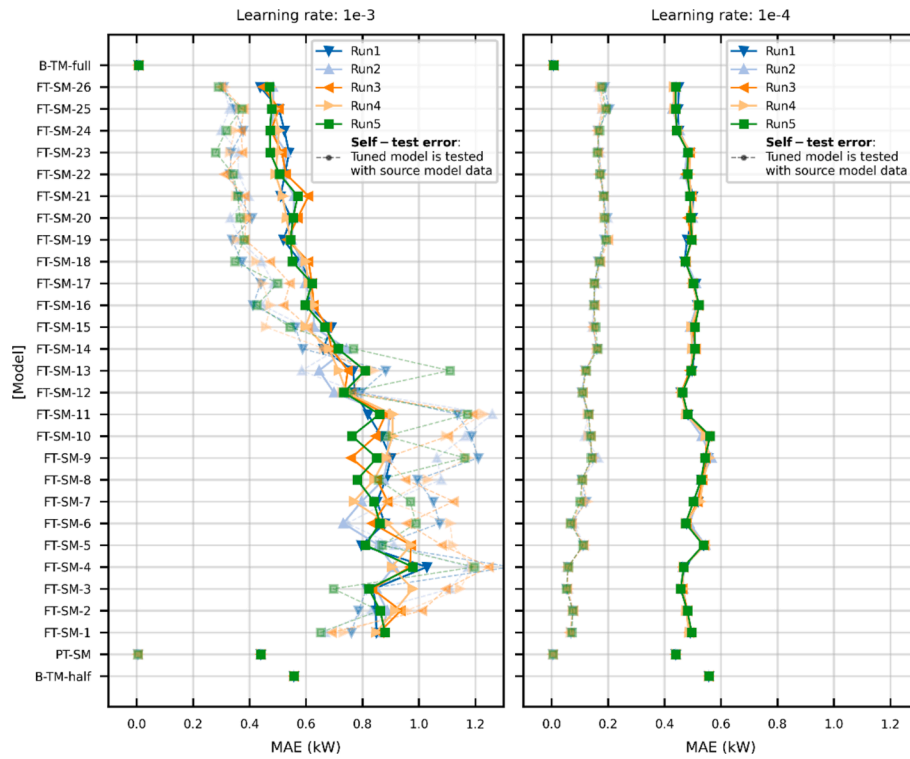


Fig. 25. Results from transfer learning repeated five times with TL-1 for two distinct learning rates ($1e-3$ and $1e-4$), the Adam optimizer and batch size of 512, with M-0_Tromsø as the source building and M-2_Bergen as the target building.

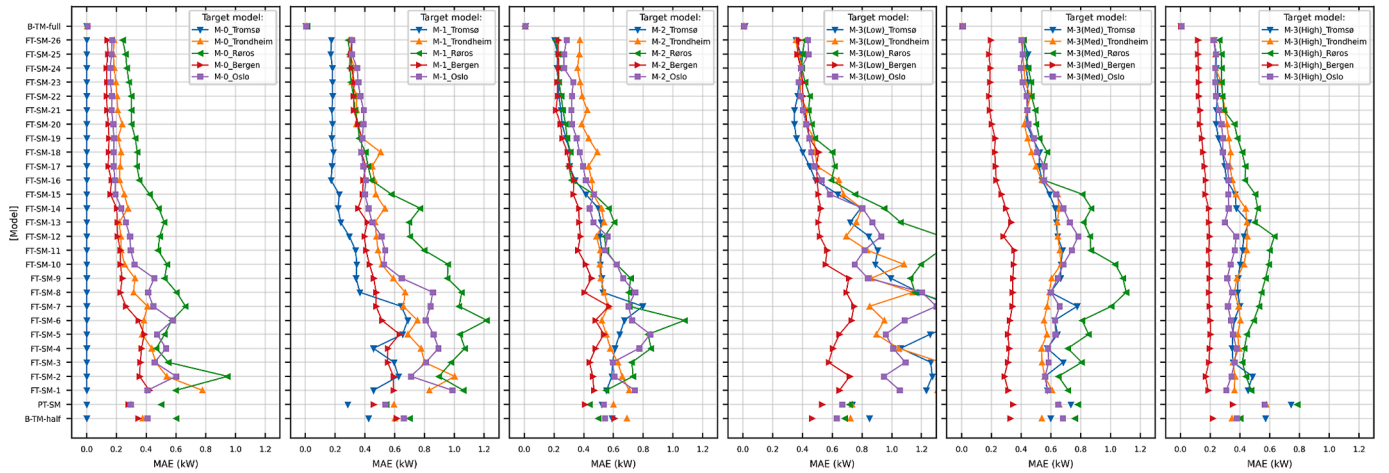


Fig. 26. Transfer-learning results for all target-building models with M-0_Tromsø as the source-building.

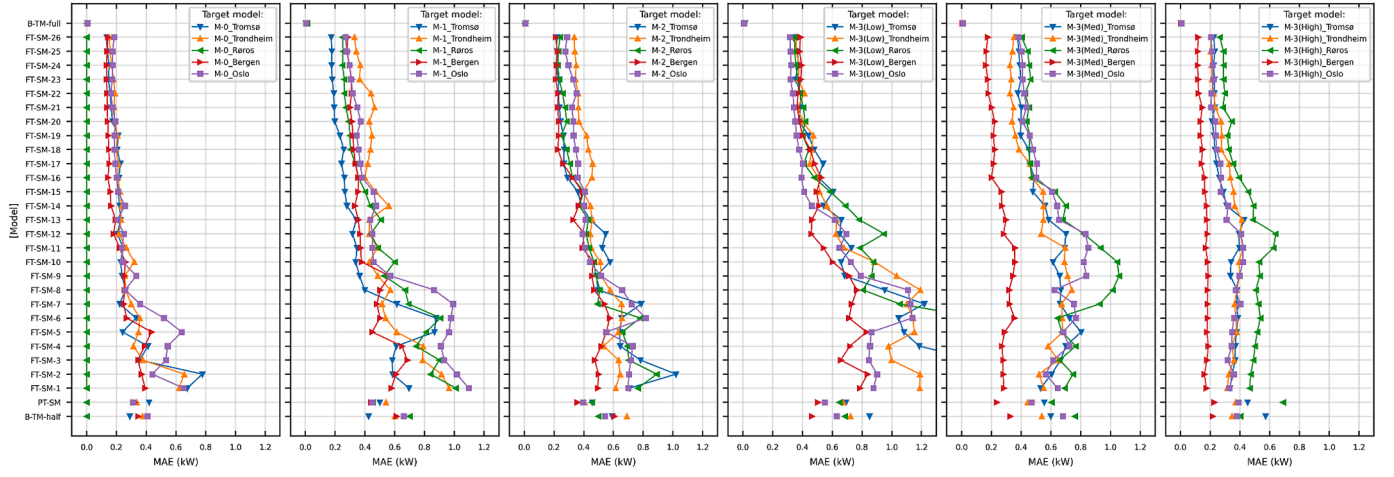


Fig. 27. Transfer learning results for all target-building models with M-0_Roros as the source-building.

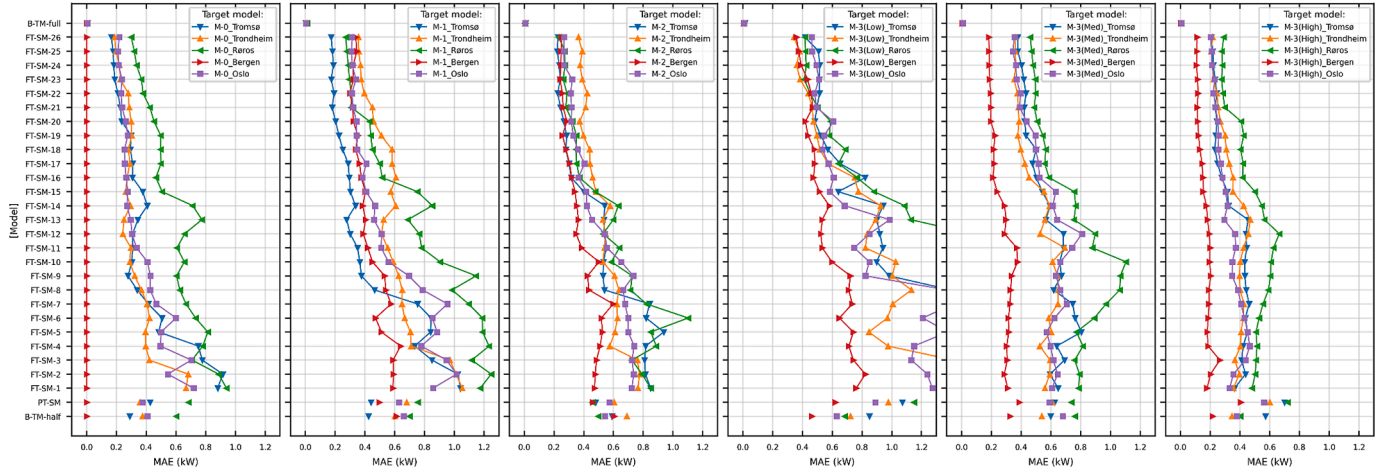


Fig. 28. Transfer learning results for all target-building models with M-0_Bergen as the source-building.

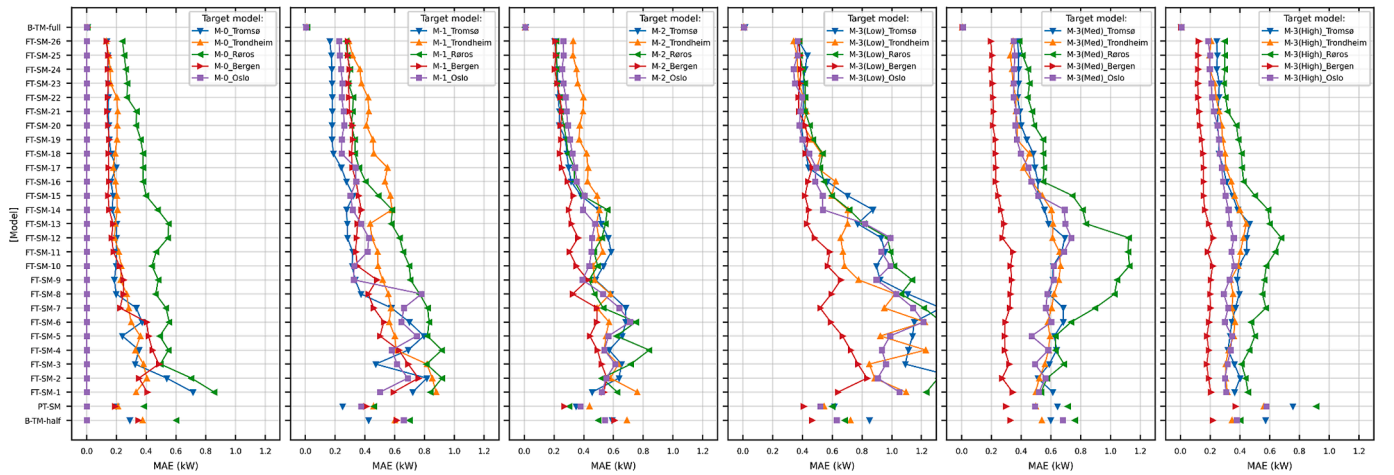


Fig. 29. Transfer learning results for all target-building models with M-0_Oslo as the source-building.

Table 5

Highest and lowest KPI values for target cities grouped by source model city and target building model.

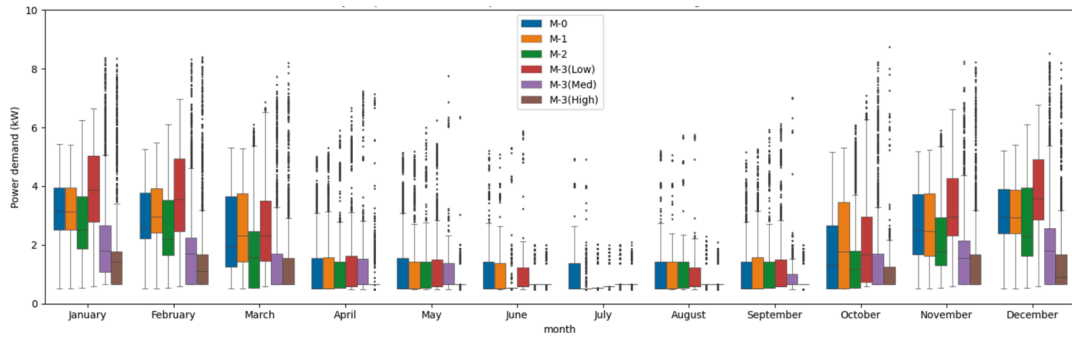


Table 6

The narrowest and broadest variability in KPI values for target cities grouped by source model city and target building model.

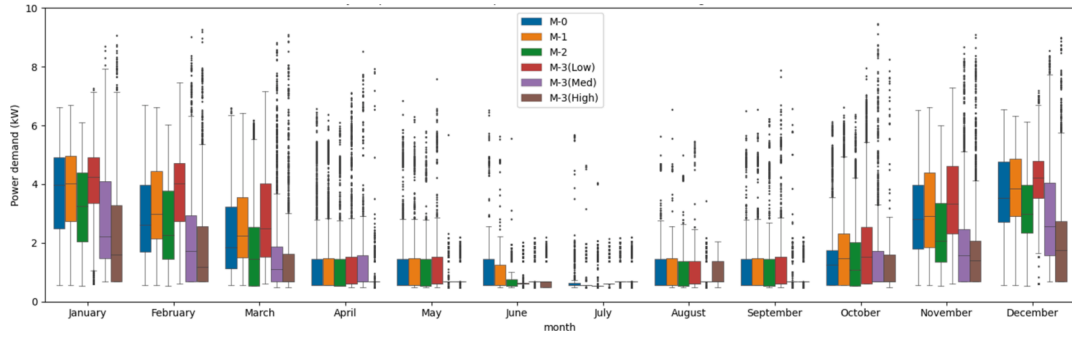
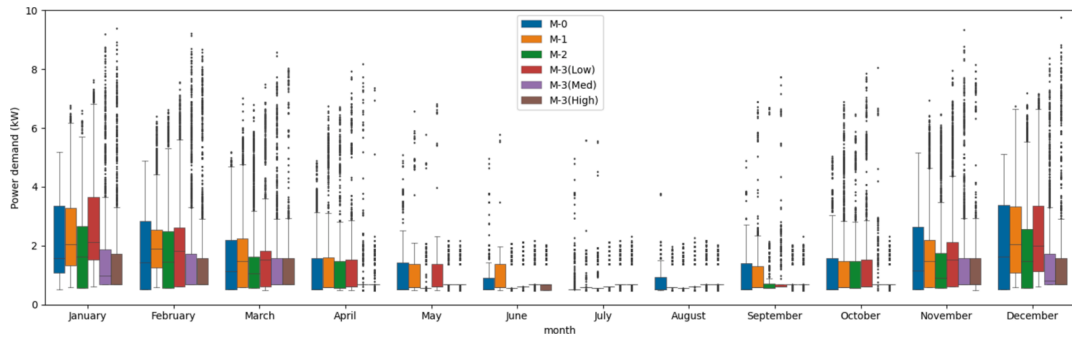


Table 7

Top 5 and bottom 5 variability ranges in KPI values for all 30 target models grouped by source model city.



Appendix B

Hierarchical data summarization used in this study.

In deep learning research, particularly in this study, the generation of vast and intricate datasets has led to results that are multi-dimensional and potentially overwhelming. In response to the challenge posed by this wealth of data, a structured and layered approach to data summarization has been employed. This approach not only offers a comprehensive view of model performance, from individual predictions to aggregated model metrics, but also facilitates the extraction of meaningful insight. While ascending through the hierarchy sacrifices some granular details, it compensates by introducing broader, overarching information. This ensures that specific data nuances might be distilled while dominant patterns and trends remain discernible. The hierarchical data summarization comprises five distinct levels, each successively refining the level of data detail and emphasizing its broader context. To encapsulate this methodology, the accompanying Fig. 30 emphasizes this hierarchical evolution.

- **Level 1** captures the detailed dynamics between the actual outputs and their corresponding Nth step-ahead predictions for the entire testing phase. To clarify, it considers the visualization of the 48th step-ahead prediction spanning a single month within the test period. This detailed representation underscores the intricacies of individual model forecasts.
- Moving a step higher, **Level 2** focuses on a quantitative metric — the Mean Absolute Error (MAE). At this point, the emphasis shifts from visual comparison to an error-based assessment between all actual outputs and their Nth step-ahead predictions throughout the testing span. A highlight on this level could be the MAE computation for a model like *FT-SM-16*, offering a numeric measure of its prediction accuracy.
- **Level 3** brings forth a visual summarization of errors. MAE values from the previous level pertaining to a specific model are portrayed via a boxplot. This visual representation, repeated for each of the 29 models in the '29-Model Bulk Evaluation Iteration', offers an insight into error distributions for a specified set of hyperparameters. An example could be the array of boxplots for all models, given a configuration like Optimizer: Adam, Learning Rate: 1-e3, Batch Size: 512, complemented with a Fine-Tuning Strategy such as *TL-2*.
- Progressing to **Level 4**, the lens zooms out further. Rather than error distributions, the focus here is on discerning trends. Each boxplot from Level 3 changes into a singular mean value of MAEs. For the models *FT-SM-1* through *-26*, these mean values are connected, sketching a trajectory that outlines the error trend. Such a view can be replicated for the entire '29-Model Bulk Evaluation Iteration', facilitating comparisons across varied hyperparameters. A similar evaluation might highlight how error trends adapt with fluctuating learning rates.
- **Level 5** summarizes the prior steps to present the transfer learning Key Performance Indicator (KPI) as described in subsection 2.5, for each fine-tuned model, specifically the *FT-SM-i* models with 'i' spanning from 1 to 26. Visually, the marker's size and opacity escalate with the value of 'i', thereby providing a subtle visual cue. This final visualization layer effectively captures variations in the transfer learning KPI across multiple hyperparameters or categories, like differing source models, amalgamating them into a unified plot.

Navigating the complexities of deep learning outcomes necessitates a strategy that balances the richness of detailed data with the clarity of high-level insights. The hierarchical data summarization method employed in this study serves as a bridge between these two requirements. It offers researchers the flexibility to delve into specifics when needed and to zoom out for an overarching perspective at other times. This method addresses the challenge of information overload, ensuring that while a plethora of data points are generated, they are never presented in a manner that obscures their meaning. By structuring the data in a layered manner, it becomes easier to trace the evolution of a single data point as it contributes to a wider trend or pattern. Furthermore, the structured approach facilitates clearer communication of findings, enabling stakeholders, from domain experts to novices, to grasp the essence of the research. The inherent scalability of this method allows for its applicability beyond the scope of this study, proving useful for any deep learning research that produces voluminous results.

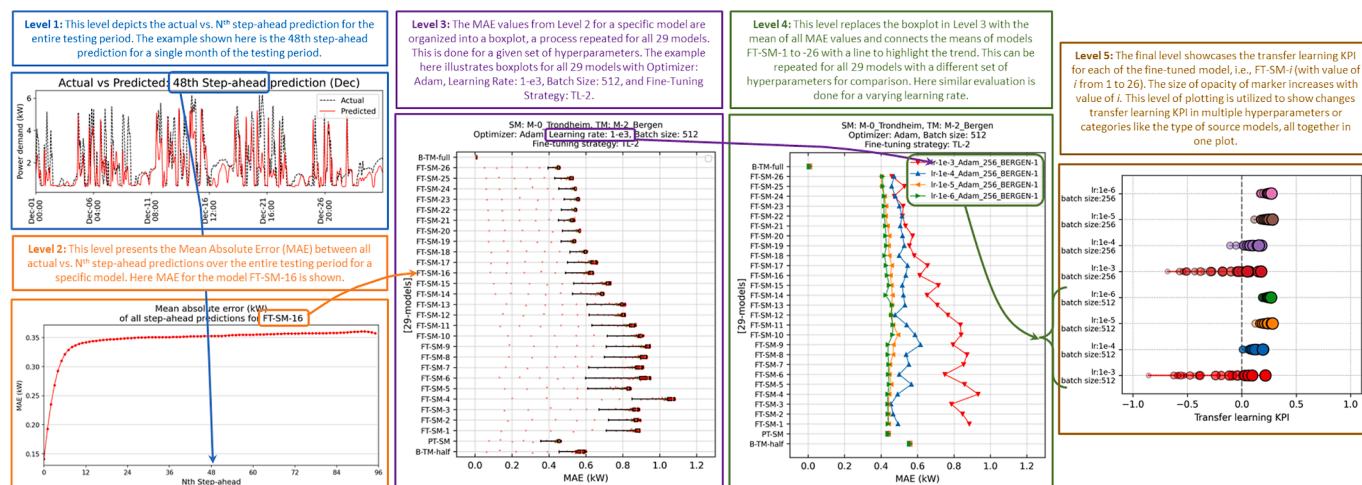


Fig. 30. Illustration of Hierarchical Data Summarization levels. The diagram showcases a step-by-step progression from detailed prediction comparisons (Level 1) to error metric computations (Level 2), followed by model-specific error distribution (Level 3), trends in error means across models (Level 4), and finally the aggregate overview of all model performance through mean error metrics (Level 5).

Data availability

Data will be made available on request.

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